



INSTALLATION OF HIGH EFFICIENCY WOOD BURNING COOKSTOVES IN LAOS-PROJECT 2



C-Quest Capital SGT Asia Stoves Private Limited

Project Title	Installation of High Efficiency Wood Burning Cookstoves in Laos-Project 2
Version	2.1
Date of Issue	23-March-2024
Prepared By	C-Quest Capital SGT Asia Stoves Private Limited
Contact	Address: C-Quest Capital SGT Asia Stoves Private Limited, 38 Beach Road #29-11, South Beach Tower, Singapore 189767 Telephone: +1-202 247-7976 Email: cgc-operations@cquestcapital.com Website: www.cquestcapital.com

CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Project	4
1.2	Sectoral Scope and Project Type	5
1.3	Project Eligibility	5
1.4	Project Design	7
1.5	Project Proponent	9
1.6	Other Entities Involved in the Project	10
1.7	Ownership.....	10
1.8	Project Start Date	10
1.9	Project Crediting Period	10
1.10	Project Scale and Estimated GHG Emission Reductions or Removals	10
1.11	Description of the Project Activity	11
1.12	Project Location	14
1.13	Conditions Prior to Project Initiation	15
1.14	Compliance with Laws, Statutes and Other Regulatory Frameworks.....	16
1.15	Participation under Other GHG Programs	16
1.16	Other Forms of Credit.....	16
1.17	Sustainable Development Contributions	18
1.18	Additional Information Relevant to the Project	18
2	SAFEGUARDS	19
2.1	No Net Harm	19
2.2	Local Stakeholder Consultation	19
2.3	Environmental Impact	21
2.4	Public Comments	21
2.5	AFOLU-Specific Safeguards	21
3	APPLICATION OF METHODOLOGY.....	21
3.1	Title and Reference of Methodology	21
3.2	Applicability of Methodology	22
3.3	Project Boundary	25
3.4	Baseline Scenario	26

3.5	Additionality	27
3.6	Methodology Deviations	28
4	QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	28
4.1	Baseline Emissions	28
4.2	Project Emissions	28
4.3	Leakage.....	29
4.4	Net GHG Emission Reductions and Removals	29
5	MONITORING	39
5.1	Data and Parameters Available at Validation	39
5.2	Data and Parameters Monitored.....	41
5.3	Monitoring Plan	46
APPENDIX 1:		58
APPENDIX 2:		58

1 PROJECT DETAILS

1.1 Summary Description of the Project

A summary description of the technologies/ measures to be implemented by the project.

The project status is under development. The project involves distribution of fuel-efficient improved cookstoves (ICS) in Laos. The ICS disseminated through this project will replace the baseline cookstoves i.e., open fire or three stone-fire or tripods using firewood or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney.

Through this project, the distribution and installation of approximately 500,000 ICS will be undertaken for 250,000 households (HHs) in Laos. It is intended that under this project two types of single pot portable stove will be distributed, which are SSM S32-X stove and Naga cookstove or any other similar ICS model may be added based on the requirement of end users as suitable. Each household will receive up to 2 ICS¹; it could comprise of different types of ICS.

The end user will be informed in advance that the use of ICS generates carbon credits which in turn will be used for subsidising the price of ICS and for recovering project implementation costs.

The location of the project.

The project will take place in Laos. The details of the project location are provided in Section 1.12.

An explanation of how the project is expected to generate GHG emission reductions or removals.

The ICS will substitute the currently practice of cooking on open fire or three stone fired stoves or a conventional device with no improved combustion air supply or flue gas ventilation that is without a grate or a chimney. The ICS burns wood more efficiently thereby improving thermal transfer to pots, hence saving fuel and lowering greenhouse gas emissions. This not only will reduce the rapidly progressing deforestation in Laos but will also reduce health hazards from indoor air pollution and women and children will have to spend less time collecting firewood.

A brief description of the scenario existing prior to the implementation of the project.

The baseline scenario is the continued use of non-renewable wood fuel by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

¹ CQC has implemented ICS projects with single ICS for HHs in its previous projects (PoA 9007, PoA 9558, PoA 8060), where it was observed that a single stove is insufficient for the household requirements and the end users continued to use the traditional stoves along with the project ICS. This led to a more practical and sustainable two-stove distribution initiative per household by CQC. This approach will ensure that the end users have complete access to clean cooking solutions and that the devices are sufficient for their cooking requirements, ensuring that the household doesn't go back to the baseline practice of three-stone fired/tripod stoves.

Even recent projects implemented by CQC (VCS 2521, 2522, 2523) have adopted the same approach followed by the current project activity too.

An estimate of annual average and total GHG emission reductions and removals.

The annual average GHG emission reduction from this grouped project is expected to be 1,223,363 tCO₂e. The total GHG emission reductions achieved for 10 years fixed crediting period is expected to be 12,233,630t CO₂e.

Project Deployment Plan

ID	Country	Deployment Plan (HHs) ²				
Year		17-February-2024 to 31-December-2024	01-January-2025 to 31-December-2025	01-January-2026 to 31-December-2026	01-January-2027 to 31-December-2027-	Total
4314	Lao PDR	39,000	72,000	72,000	,67,000	250,000

Audit History

Audit Type	Period	Program	VVB Name	Number of years
Validation	31-March-2023 to 24-June-2023 ³	VCS	Carbon Check (India) Private Limited	<1 year(86 days)

1.2 Sectoral Scope and Project Type

The project is categorised under type/category as below:

- a) **Sectoral scope**: 03 - Energy demand
- b) **Type**: II – Energy efficiency improvement projects.

The project is a non-AFOLU grouped project.

1.3 Project Eligibility

As per para. 2.1.1 of the VCS Standard, V.4.4, the project is eligible under the scope of the VCS Program because the VCS Scope includes project activities involving the following elements:

Table 1: VCS Programme Scope

² As part of this grouped project 2985 SSM S-32X and 2985 Naga cookstoves were distributed in the Lao PDR as of 29-February- 2024.

³ Appointment of VVB and issue of Validation report by VVB. Including both days under number of years

Scope	Applicability
1 The six Kyoto Protocol greenhouse gases.	The project activity reduces GHG emissions associated with the combustion of cooking fuels. The project activity aims at CO ₂ reduction of non-renewable biomass.
2 Ozone-depleting substances.	The project does not involve any ozone depleting substances as project devices use fuelwood in the ICS.
3 Project activities supported by a methodology approved under the VCS Program through the methodology approval process.	The project activity is distribution of improved cookstoves, "VMR0006 "Methodology for Installation of High Efficiency Firewood Cookstoves", version 1.1." which is a VCS approved methodology and is applicable for the project activity.
4 Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	Not applicable.
5 Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements.	Not applicable.
As per para 2.1.2 of the VCS Standard 4.4	
Scope of the VCS Program excludes projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction.	Not applicable

Project stoves i.e., ICS, fall under the category of efficiency improvements in thermal applications, therefore eligible under the scope of VCS Programme. Also, the project activity efficiency improvements in thermal applications e.g., cook stoves is not listed in the category of excluded project activities table in section 2.1.3 of the VCS Standard ver. 4.4, and therefore, the project activity is eligible under the scope of the VCS Program.

1.4 Project Design

- ☐ The project includes a single location or installation only.
- ☐ The project includes multiple locations or project activity instances but is not being developed as a grouped project.
- ☒ The project is a grouped project.

The project is a grouped project. In this grouped project, each ICS constitutes a single project activity instance which is in line with its definition stated in VCS Program Definitions version 4.3.

Eligibility Criteria

For the inclusion of new project activity instances, the project proponent shall ensure that it meets the eligibility criteria according to section 3.6.16 and 3.6.17 of the VCS Standard 4.4 as stated in the below table.

Table 2: lists the eligibility criteria according to VCS Standard 4.4

S.No.	Criterion	New project activity instances -Compliance
1	Meet the applicability conditions set out in the methodology applied to the project.	New project activity instances, Improved cookstoves (each SSM-S32-X Stove, Naga Cookstove, or other model stoves) will meet the applicability conditions set out in Section 3.2 where the target of the end-user is household.
2	Use the technologies or measures specified in the project description.	SSM-S32-X Stoves and Naga Cookstoves to be adopted in the project, (other ICS models may be added) having minimum efficiency of 25%; below which it shall not be eligible under this grouped project
3	Apply the technologies or measures in the same manner as specified in the project description.	SSM-S32-X Stoves and Naga Cookstoves to be adopted in the project and it will replace traditional cookstoves in household
4	Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	The new project activity instances will be distributed within Laos only and subject to the same baseline scenario determined in Section 3.4.
5	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.	All new project activity instances will use the activity method for demonstration of additionality. Step 1: Regulatory Surplus There is no mandated government Programme or policy in the host country of this project ensuring the distribution of new project activity instances.

		Step 2: Positive List 1. The inclusion of new project activity instances shall have to comply with positive list that is the ICS are installed/ distributed at zero cost to end user and the project activity instance has no other source of revenue but GHG credits. 2. Project activities that are implemented as part of government schemes or are supported by multilateral funds cannot be considered additional even if the stoves are distributed free of cost or at a highly subsidized rate and hence are not eligible to use this methodology. Step 3: Project Method For any project activity where stoves are not provided at zero cost to the end-user or has any other source of revenues other than the sale of GHG credits, then the project activity shall apply investment analysis method set out in the CDM Tool for the Demonstration and Assessment of Additionality included in AMS II.G to determine that the proposed project activity is either. 1) not the most economically or financially attractive, or 2) not economically or financially feasible.
6	Ownership: Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance.	A default Beneficiary Agreement for end users including the provision that emission reductions generated by the project activity are owned by the Project Proponents will be provided for project activity instance.
7	The start date of project activity instance shall be same as or later than the grouped project start date.	Project Activity Instance Database containing date of installation of each ICS.
8	Where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit. Further, no single cluster of project activity instances shall exceed the capacity limit, determined as follows: 1) Each project activity instance that exceeds one percent of the capacity limit shall be identified. 2) Such instances shall be divided into clusters, whereby each cluster is comprised of any system of instances such that each instance is within one	No project activity instance shall exceed the applicable limit, which is 180 GWh_{th}/y. The expected annual energy saving for each project activity instance is approximately 0.018 GWh_{th}/y or 0.010% of the limit for SSM S-32X cookstove and the expected annual energy saving for each project activity is 0.0068 GWh_{th}/y or 0.004% of the limit for Naga cookstove. As the annual energy saving is below 1% of the limit, no project activity instance is identified and divided into clusters.

	kilometre of at least one other instance in the cluster. Instances that are not within one kilometre of any other instance shall not be assigned to clusters. 3) None of the clusters shall exceed the capacity limit and no further project activity instances shall be added to the project that would cause any of the clusters to exceed the capacity limit.	
9	Double counting: The Improved Cookstove distributed under this project shall be uniquely identifiable based on the distribution records.	Project Activity Instance Database containing unique serial number of each ICS. Each ICS distributed will have corresponding end user (Stove owner) details (i.e., name, geographical coordinates, address, unique serial number). The combination of this unique serial number of each ICS along with geographical address of the stove owner confirms the unique identity of each project device.
10	Target Population: End user for each project activity instance shall be households, with non- renewable biomass on inefficient firewood stoves.	Each ICS will be assigned a unique serial number with name of ICS user, address, GPS of household, stove model, distribution date, etc.,
11	Crediting period: Be eligible for crediting from the start date of the instance through to the end of the project crediting period (only). Note that where a new project activity instance starts in a previous verification period, no credit may be sought for GHG emission reductions or removals generated during a previous verification period (as set out in Section 3.4) and new instances are eligible for crediting from the start of the next verification period.	The start date of the instance will be the date the stove began generating GHG emission reductions or removals. The end of the crediting period of instance will be the end date of project crediting period as mentioned in section 1.9.

1.5 Project Proponent

Organization name	C-Quest Capital SGT Asia Stoves Private Limited
Contact person	Ken Newcombe
Title	Director

Address	38 Beach Road #29-11, South Beach Tower, Singapore 189767
Telephone	+1-202 247-7976
Email	cqc-operations@cquestcapital.com

1.6 Other Entities Involved in the Project

C-Quest Capital SGT Asia Stoves Private Limited is the sole entity involved in the project.

1.7 Ownership

During the handover and registration of ICS, the participating household will sign a declaration to transfer the ownership rights of the carbon assets generated from this project to the project proponent.

Further, records of consent letter/ record of consent signed by end user (ICS owner) to PP during the registration process includes the statutory clause that " household hereby transfers and waives all and any rights in connection with greenhouse gas emission reductions or other environmental or social attributes generated by the installation and use of the improved cookstove to C-Quest Capital SGT Asia Stoves Private Limited acting as project manager on behalf, and for the benefit of its principals and we understand that C-Quest Capital SGT Asia Stoves Private Limited may, at any time, transfer those rights to one of its principals without further notice or consent." This is in line with clause 5 of section 3.7.1 of VCS Standard v4.4.

1.8 Project Start Date

17-February-2024 (Date of distribution of the 1st ICS under this grouped project activity).

1.9 Project Crediting Period

17-February--2024 to 16-February-2034, 10 years fixed crediting period

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☐ <20,000 tCO₂e/year
- ☐ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year
- ☒ >1,000,000 tCO₂e/year

Project Scale	
Project	
Large project	X

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
17-February-2024 to 31-December-2024	315,011
01-January-2025 to 31-December-2025	803,247
01-January-2026 to 31-December-2026	1,272,029
01-January-2027 to 31-December-2027	1,622,135
01-January-2028 to 31-December-2028	1,573,224
01-January-2029 to 31-December-2029	1,474,762
01-January-2030 to 31-December-2030	1,382,402
01-January-2031 to 31-December-2031	1,295,768
01-January-2032 to 31-December-2032	1,214,507
01-January-2033 to 31-December-2033	1,138,290
01-January-2034 to 16-February-2034	142,255
Total estimated ERs	12,233,630
Total number of crediting years	10
Average annual ERs	1,223,363

1.11 Description of the Project Activity

The households will receive an ICS as an energy efficiency measure. ICS will be distributed to urban and rural households where it will be used for daily cooking in replacement of traditional wood stoves. Households may receive more than one ICS, depending on their size and cooking habits.

Introduction of ICS as energy efficiency measure:

The project will distribute ICS to replace existing devices. The ICS will continue to consume woody biomass (firewood) for cooking. The ICS will consume less wood fuel as it has higher thermal efficiency, and it would result in a reduction of GHG emissions from the baseline scenario.

Technology/Measure

It is envisaged that the ICS models to be deployed in this project will be SSM-S32-X and locally produced Naga Stove. However, with the availability of new technologies, the PP may consider deploying other models as well. The stoves will substantially reduce woody biomass fuel consumption and emissions during cooking.

SSM-S32-X



Figure 1: SSM S-32X

Table-3 Technical description of SSM S-32X⁴

No.	Parameter	Value
1	Thermal Efficiency	51.50% (with pot skirt)
2	Dimensions	25.5 cm height, 32 cm diameter
3	Material	• Stove Body: Stainless Steel • Pot Rest: Cast Iron • Pot Skirt: Stainless Steel • Burning Chamber: Stainless Steel and Ceramic • Fuel Feeding Door: Stainless Steel • Inner Fuel Shelf: Stainless Steel • Outer Fuel Shelf: Powder-coated steel
4	Life Span	10 years

Naga Cookstove

Table 4: Technical description of Naga Cookstove⁵

⁴ Manufacturer Specification

⁵ Manufacturer Specification

No.	Parameter	Value
1	Thermal Efficiency	35%
2	Dimensions	28 cm height (+/- 0.5 cm), 33 cm diameter (+/- 0.5 cm),
3	Material	Combustion Chamber and Stove Top: Ceramic made from locally available clay. Stove Body: Aluminum Sheet Metal
4	Life Span	More than 3 years

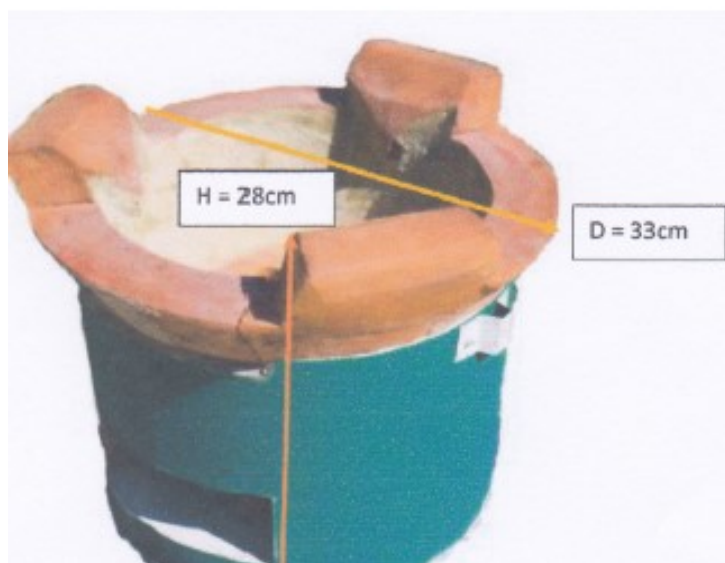


Figure 2: Naga Cookstove

Maintenance works such as repairs, replacement of stove parts as well as replacement of stoves, will be carried out as and when required to ensure that the stoves remain operational and are in good condition throughout the crediting period of 10 years.

Data collection of ICS end-user

Project proponent will gather the necessary information to identify households using their ICS during the project. To facilitate this process, each ICS will be assigned a unique serial number. This number will be recorded during the registration process together with the following information (as appropriate and as available).

- Name of ICS user or head of the household.
- Address / GPS of ICS household.
- Phone number of ICS user or household, where available.
- Stove model.
- Date of distribution/installation.
- ICS serial number.

- Retailer/distributor information.

The information collected will be stored in the electronic database which will serve as the project database for project monitoring and sampling purposes.

In case a replacement stove is being issued to a customer already registered on the project database, a new registration will not be required. The replacement stove will be recorded in the project database in such a way that the replaced stove ceases to be included in the project as a new registration.

1.12 Project Location

The project location will be the geographical boundary of Lao PDR with coordinates 19° 51' 22.6" N latitude and 102° 29.73' E longitude.⁶



Figure 3: Laos map⁷



Figure 4: Map of project area

Laos is divided into 17 provinces and one prefecture, the Vientiane capital city municipality. The provinces are grouped geographically into 3 stratas, North (from Phongsaly to Saiyabouly, Luang Prabang and Xiengkhuang), Central (Vientiane and Bolikhamxay) and South (from Khammuane to Champasack). To facilitate the management, implementation, monitoring and sampling stages of the project, the project proponent has divided the project boundary into three project regions according to the location of provinces.

⁶ <https://www.geodatos.net/en/coordinates/laos>

⁷ <https://www.mapsofworld.com/laos/laos-political-map.html>

Table 5: Location of the project

No.	Project regions	Provinces
1	North	Houaphanh, Luang Namtha, Luang Prabang, Oudomxay, Phongsaly, Singable, Xaisomboun, Xiangkhouang, Bokeo.
2	Central	Vientiane, Vientiane Capital, Bolikhamxai.
3	South	Attapeu, Champasak, Khammouane, Salavan Savannakhet, Sekong.

The geodetic coordinates and KML file⁸ for first project instances of grouped project activity and geodetic polygons to delineate the project's geographic area provided to VVB and the location of the first instance as shown below:



Figure 5: First instance geodetic coordinate with the geodetic polygons to delineate the project's geographic area.

1.13 Conditions Prior to Project Initiation

The condition prior to project implementation is the continued use of three stone fired cookstoves or a conventional device with no improved combustion air supply or flue gas ventilation (that is without a grate or a chimney) by the target population by burning non-renewable woody biomass

⁸ KML file provided to VVB

to meet the thermal energy needs as provided by project cookstoves in absence of project activity. Also, more information is provided in section 3.4 below.

It is a common and traditional practice in the project area to use three stone fired stoves or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney. Hence, the applicable efficiency fraction for three stone fired or conventional stoves which stands at 0.1 (default) as per the applied methodology VMR0006 1.1 is considered in the baseline estimations.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

There are no laws and regulations governing the use of improved cookstoves in Laos households. The project is a voluntary effort by the project proponent.

A review is made on the main legislation below:

Law on Environmental Protection and Natural Resource Management

- to protect [and] promote environmental quality and public health through the prevention, reduction, and control of pollution.
- to assess the environmental impact of all proposed projects prior to the issuance of a decision by the Royal Government.
- to ensure the rational and sustainable conservation, development, management, and use of the natural resources of the Kingdom of Laos.
- to encourage and enable the public to participate in environmental protection and natural resource management.
- to suppress any acts that cause harm to the environment.

There is no negative concern made on improved cookstoves project from the above law.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not been registered, nor is it seeking registration under any other GHG program.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes

☒ No

The project is not included in an emissions trading program or any other mechanism that includes GHG allowance trading.

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes ☒ No

The project has not sought or received another form of GHG-related environmental credit.

Supply Chain (Scope 3) Emissions

Have the owner(s) or retailer(s) of the impacted goods and services⁹ posted a public statement saying, “VCUs may be issued for the greenhouse gas emission reductions and removals associated with [Installation of High Efficiency Wood Burning Cookstoves in Laos-Project 2] [C-Quest Capital SGT Asia Stoves Private Limited]” since the project’s start date?

☒ Yes ☐ No

Has the project proponent posted a public statement saying, “VCUs may be issued for the greenhouse gas emission reductions and removals associated with [Installation of High Efficiency Wood Burning Cookstoves in Laos-Project 2][Lao PDR, C-Quest Capital SGT Asia Stoves Private Limited].”

☒ Yes ☐ No

Have the producer(s) or retailer(s) of the impacted good or service been notified of the project and the potential risk of Scope 3 emissions double claiming via email?

☒ Yes ☐ No sign

As per Section 3.23.9 of the VCS Standard, v4.4, the “producer(s) or retailer(s) of the impacted good or service are known but not involved in the project or do not have a website”. The Project Proponent has informed the manufacturers of ICS as well as the implementing partner in Lao PDR at the time of signing agreement for the goods and services provided by them, that ICS will be distributed in Lao PDR under this project activity. Also, intimated them about the project registration under VERRA for the generation of VCUs.

A public statement(s) by the Project Proponent has been made throughout the project crediting period on its website for reference. The impacted good or service's producer(s) have been notified of the project and the potential risk of Scope 3 emissions double claiming via email. Evidence of the public statement(s) and email(s) have been provided to the VVB for validation process.

⁹ Impacted goods and services are all goods and services directly impacted by the technologies and measures specified as project activities in the project description. Please see the VCS Program document *VCS Program Definitions* for additional information.

1.17 Sustainable Development Contributions

A clean cookstove project results in a number of socioeconomic benefits to the society in addition to environmental benefits.

The project will contribute to the following Sustainable Development goals such as

- SDG 3: Good Health and Well-being
- SDG 4: Quality Education
- SDG 5: Gender Equality
- SDG 7: Affordable and Clean Energy
- SDG 8: Decent work and economic growth
- SDG 13: Climate action, and
- SDG 15: Life on Land

The contributions to the social, environmental and economic goals mentioned above is achieved in the following ways.

a) Environmental Sustainability

- The project will help significantly reduce greenhouse gas emissions over its lifetime. (SDG-13)
- The project will help reduce the use of non-renewable biomass from forests, thus will assist in conserving existing forest stock and the protection of natural forest eco-systems and wildlife habitats (SDG-15)

b) Social Sustainability

- The project stoves will considerably reduce the time that need to be spent in collecting wood fuel and cooking activities of the household thereby reducing the work burden on rural families and presenting alternative opportunities for economic development. Also, it will avoid the need for children from spending time for fuel wood collection and supporting family in cooking meals, hence more time available for them for focusing on quality education (SDG-5) (SDG-4)
- The number of indoor pollutants from the burning of woody biomass in the household will be reduced. Less carbon dioxide, carbon monoxide and particulates will be emitted due to the decrease in total woody biomass burned (SDG-3)
- The households will have access to affordable and clean energy through the distribution and usage of ICS (SDG-7)
- The project stoves provide a safer method of burning woody biomass for cooking, thereby helping reduce burn injuries, especially for children, in the household. (SDG-3)

c) Economic Sustainability

- The project will help develop a section of the local economy, in the distribution, local assembly, maintenance and monitoring activities. (SDG-8)
- Household expenditure on cooking fuel will be reduced using the ICS. (SDG-7)
- Saved household labor can be diverted to more productive economic activities. (SDG-8)
- The employment opportunity will also equip them with skills which are expected to improve their future employment prospects. (SDG-8)

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable. The project considers a gross adjustment factor of 95% to account for leakage as per the applied methodology.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not applicable

2 SAFEGUARDS

2.1 No Net Harm

The project will only bring positive impacts on environmental and socio-economic aspects as elaborated in Section 1.17. No potential negative environmental or socio-economic impacts have been identified for the project.

2.2 Local Stakeholder Consultation

Local Stakeholders Consultation has been undertaken at Project Level. Local Stakeholder Consultation are key components of the project operations. Project Proponent and Distribution Partners (on behalf of PP) undertake interactive training and demonstration sessions with community members who receive clean cookstoves. These sessions provide information on how to build/assemble, operate and area for the ICS as well as to engage with end users regarding queries that they may have in relation to the use of stoves.

The procedures or methods of conducting LSC and regular feedback mechanism are.

Local Stakeholders Consultation Webinar	The local stakeholder consultation webinar for the Grouped Project Activity- Installation of High Efficiency Wood Burning Cookstoves in Laos was conducted on 28-April-2021 from 10:00 to 12:00 hours (Lao PDR Local Time).
Mode of Invitation and identified stakeholders	The concerned stakeholders were informed about the webinar across various platforms including advertisements in national newspapers, personal mails sent to identified stakeholders such as probable beneficiaries, local NGOs working in the

	field of energy efficiency and women empowerment, officials from Ministry Of Natural Resources and Environment, Ministry of Energy and Mines, Ministry of Agriculture and Forestry, various government officials responsible for rural development and clean energy, educationists, international agencies working in the field of clean cooking etc. Social media platforms such as Facebook, LinkedIn and WhatsApp were also used for broadcasting information about the webinar and inviting concerned stakeholders. Information about the webinar was also posted on C-Quest Capital website.
Response Mechanism	The stakeholders were requested to register their response through evaluation form (sample attached in appendix), or they could send their comments at the mail id provided or on the link made available at the project proponent 's website.
Outcome of LSC meeting	The outcome of the LSC meeting was more than encouraging. By the end of the webinar and even after, several stakeholders expressed their desire to connect with this Programme as they truly felt that it could be a catalyst in bringing about household energy efficiency improvements in the rural and financially backward sections of society as well as preserve our pristine forests. None of the queries or comments raised by stakeholders (attached in Appendix), required any design change, albeit the proponents did realize that the suggestions put forth by stakeholders would go a long way in ensuring uptake and long-term use of the technology by the end users. Further the queries have been addressed and response made during the consultation were prepared as a LSC report (provided the same to VVB as a part of validation). Stakeholders were also given an evaluation form to express their feedbacks about their projects. The questions raised by the stakeholders through the feedback form were also addressed.

On-going consultation mechanism	The project proponent will have an on- ground team to collect continuous feedback from end users as these are the most important stakeholders who ultimately decide the success of an intervention. In addition, the stakeholders can directly contact the project proponent via call or e-mail or letter. These details are shared with the end user at the time of project registration. End Users can also report their grievances and complaints regarding stove parts via calling on the helpline number which has been engraved on the ICS serial number plate.
---------------------------------	---

2.3 Environmental Impact

No negative environmental impacts have been identified from the project and environmental impact assessment (EIA) is not required for the project.

2.4 Public Comments

For the global stakeholder consultation, the VCS-PD was hosted on the VERRA website for public comments from 20-April-2023 to 20-May-2023. No public comment was received during this period.

2.5 AFOLU-Specific Safeguards

This section is not applicable as the project is a non-AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1¹⁰

In addition, the following tools and standards have been used.

- Tool 30: “Calculation of the fraction of non-renewable biomass” (version 4.0¹¹)

¹⁰<https://verra.org/wp-content/uploads/2021/07/VMR0006-Methodology-for-Installation-of-High-Efficiency-Firewood-Cookstoves-v1.1.pdf>

¹¹ <https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-30-v4.0.pdf>

- Standard: “Sampling and Surveys for CDM project activities and Programme of activities” (version 09.0¹²)

3.2 Applicability of Methodology

VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1

Applicability criterion	How the project complies
Project activities shall be implemented in domestic premises or in community-based kitchen.	The proposed project involves deployment of ICS only in households. .
The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer’s specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	For the present project activity instances SSM-S32-X and Naga Cookstoves are envisaged to be distributed in end user households. SSM- S32-X Cookstove is a natural draft, side-feed semi-gasifier stove. The high-power thermal efficiency of stove is 51.5% with pot skirt as specified by the manufacturer. Locally produced Naga Cookstove is envisaged to be distributed in combination with SSM-S32-X, to ensure that the households do not go back to using baseline stoves. It is a single stove, artisanal made, bucket stove made from clay and metal. The high-power thermal efficiency of this stove is 35% as specified by the manufacturer.
Both ‘Projects’ and ‘Large Projects’ can use the methodology. Each cookstove shall result in annual energy saving of less than or equal to 1% of the capacity limit indicated in the methodology.	The proposed project is expected to generate 12,233,630 tonnes CO ₂ per year which is greater than 300,000 tonnes CO ₂ per year and thus qualifies as “large project” as per section 3.10.1 of the VCS Standard version 4.4. Each cookstove ¹³ results in annual energy saving of 0.018 GWh in case of SSM-S32-X which is 0.010% of the energy limit and 0.0068 GWh in

¹² https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth_Stan05.pdf

¹³ Indicative value. Actual value will depend on the type of stove installed.

	case of Naga cookstoves which is 0.004 % of the energy limit and these are below the 1% threshold set under VCS Standard, v4.4 (Paragraph 3.6.9).
Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports, or statistics;	Non-renewable biomass has been used since 31 December 1989 in Lao PDR as demonstrated at below.
For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded.	Not applicable for this project activity. The ICS is introduced as energy efficiency measure to replace baseline stoves and will reduce the use of non-renewable biomass used for cooking.
The project document shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo).	The present project activity instance plans to distribute combination of portable SSM-S32-X, and locally produced Naga Cookstoves. Other ICS models may also be distributed based on the requirement of end users. Both the stoves will bear unique identification in the form of serial number linking it clearly to the present project activity instance. Also, combination of end user 's details (i.e., name, geographical coordinates, address) along with unique serial number of the ICS maintained in the database will make the project device clearly identifiable from any other project device. Hence there will be no double counting of the

	project device within the project or from other project activity. In addition, the project will be cross checked against other CDM project activity operating in the country using the UNFCCC, the Gold Standard, and other relevant voluntary carbon schemes to ensure that the ICS is not included in any other CDM project activity or voluntary project activity.
The project document shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	An undertaking will be taken by all the beneficiaries, i.e., the end-users indicating that the ownership of the current project lies solely with C-Quest Capital SGT Asia Stoves Private Limited. To further ensure that the ownership rights of the project are reserved with CQC, all the stakeholders relevant, i.e., manufacturers/wholesale providers will be provide their consent which will clearly state that the project proponent or an entity authorized by it shall be the sole owner of the VCUs arising from the project.

Evidence that the non-renewable biomass has been in use since 1989.

Non-renewable biomass has been in use since December 31, 1989, as evidenced by various FAO statistical data. The Global Forest Resources Assessment 2020¹⁴ (FAO) indicates that forest areas decline yearly. Between years 1990 and 2020, Lao PDR lost on average of 41,583.3 hectares of forest per year. This amounts to an average annual deforestation rate of 0.24%. Between the years 2000 and 2020, the rate of forest loss is 0.23% per annum and between 2010 and 2020, the rate of forest loss is 0.20% per annum. In total, between 1990 and 2020, Lao PDR lost 6.99% of its forest cover, or around 1,247,500 hectares.

Table 6: Decline in forest area between 1990 – 2020.

FRA 2020 categories/Year	Area(1000 ha)			
	1,990	2000	2010	2020
Forest(a)	17,843.00	17,425.00	16,940.50	16,595.50
Other wooded land (a)	138.00	136	132	130
Other land (c-a-b)	5,099.00	5,519.00	6,007.50	6,354.50
Total land area (c)	23,080.00	23,080.00	23,080.00	23,080.00

Thus, it can be concluded that losses have been reported in forest areas from year 1989 thereby hinting that fuelwood used across Laos comes from non-renewable sources.

¹⁴ FAO. Global Forest Resources Assessment 2020. Country Report. Laos, Page 17

Applicability criteria of TOOL 30: Calculation of the fraction of non-renewable, Version 04.

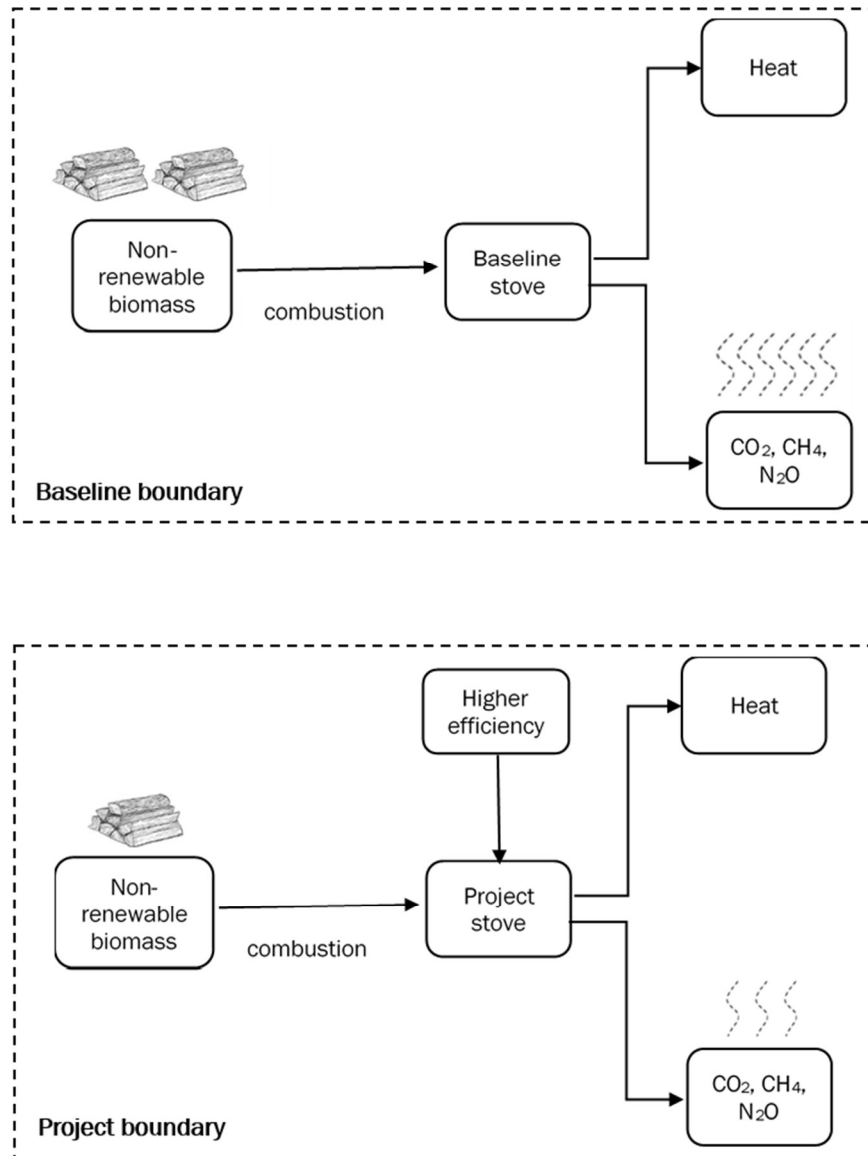
S.no	Applicability criterion	How the project complies
1.	This tool may be used by: (a) DNAs to submit region- or country-specific default f_{NRB} values, following the procedures for development, revision, clarification, and update of standardized baselines (SB procedures); or (b) project participants to calculate project- or PoA-specific f_{NRB} values.	Applicability is met through option (b) and PP engaged a third-party competent agency (C4 Eco Solutions) to calculate the country specific f_{NRB} based on Tool 30, version 04. f_{NRB} calculations have been submitted to VVB.

3.3 Project Boundary

For Improved cook stoves as energy efficiency measure, the project boundary is the physical, geographical site of the efficient devices that utilize woody biomass. The project location will be the geographical boundary of Lao PDR. .

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Yes	Major source
	CH ₄	Yes	Major source
	N ₂ O	Yes	Major source
	Other	No	No other source identified
Project	CO ₂	Yes	Major source
	CH ₄	Yes	Major source
	N ₂ O	Yes	Major source
	Other	No	No other source identified

A representation of the baseline boundary and project boundary are given as below.



3.4 Baseline Scenario

As per the methodology VMR0006 V1.1, the baseline scenario is the continued use of three stone fired cookstoves or a conventional device with no improved combustion air supply or flue gas ventilation (that is without a grate or chimney) by the target population by burning non-renewable wood fuel (firewood) to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

The project activity instances to be included in this grouped project activity should meet the baseline scenario as mentioned above.

The project activity will target households using fuelwood-fired traditional cookstoves or three-stone fire stoves in the region. Households using fuels other than firewood or those already using

improved cookstoves or cook stoves with chimney, flue gas ventilation systems, and grates will not be eligible for selection under the project activity, in accordance with the beneficiary selection criteria.

The baseline scenario can also be confirmed through the registration cum consent deed signed by the registered stove owner during the distribution of the project device (ICS). This deed confirms that the registered user voluntarily adopted the project stove and was previously using a three-stone fire or conventional device (without a grate or chimney) before switching to the project stove.

PP has conducted comprehensive literature reviews and data studies to understand the baseline scenario in Lao PDR. According to the Biomass Consumption and Survey in the Household Sector by the Ministry of Energy and Mines, Lao PDR in June 2021¹⁵, 69% of households rely on fuelwood for cooking. Specifically, 53% of urban households and 78% of rural households mainly rely on fuelwood for cooking. Further, prior to distributing/installing the Improved Cook Stoves (ICS), PP/IP representatives will ensure that the ICSs are installed in households currently using traditional cookstoves/three-stone fires or conventional cooking systems, and where wood is the primary fuel.

With continue using the traditional inefficient cooking stoves, consuming high quantity of non-renewable biomass. Thus, the baseline scenario has been identified in line with the requirements of the applied methodology VMR0006, Version 1.1

The project cookstoves (ICS) will be deployed substituting the less efficient baseline stoves. For this project as per applied methodology VMR0006 V 1.1 default value of 10% for efficiency of baseline stove i.e., three-stone cookstove using firewood has been considered.

3.5 Additionality

Methodology VMR0006 uses activity method for the demonstration of additionality.

Activity Method

Step 1: Regulatory Surplus

There is no mandated government programme or policy in host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law,

¹⁵ Establishment of Energy Statistic Regulation in Lao PDR (eria.org) (page number-7/22)
https://www.eria.org/uploads/media/Research-Project-Report/2021-05-Establishment-Lao-Energy-Statistics-Regulation/15_Ch.7-Biomass-Household.pdf

statute, or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute, or other regulatory framework.

Households may only participate voluntarily in this project. It is hereby confirmed that the proposed project is a voluntary coordinated action by CQC.

Step 2: Positive List

As per Section 3.2, the project meets the applicability conditions of the methodology which represent the positive list.

The project installs the ICS at zero cost to the household and the project is not feasible without the revenue from sale of GHG credits.

The project is not implemented as part of government schemes or supported by multilateral funds.

Conclusion: As the project fulfils the conditions above, it is deemed additional.

Step 3. Project Method

For any project activity where stoves are not provided at zero cost to the end-user or has any other source of revenues other than the sale of GHG credits, then the project activity shall apply investment analysis method set out in the CDM Tool for the Demonstration and Assessment of Additionality included in AMS-II.G to determine that the proposed project activity is either:

- 1) not the most economically or financially attractive, or
- 2) not economically or financially feasible

The project installs the ICS at zero cost to the household and has no other source of revenue other than the sale of GHG credits. Hence this condition is not applicable.

3.6 Methodology Deviations

The project did not apply any methodology deviations.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The methodology does not account for baseline emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4.

4.2 Project Emissions

The methodology does not account for project emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4.

4.3 Leakage

Leakage shall be considered as default 0.95 in accordance with methodology.

4.4 Net GHG Emission Reductions and Removals

The improved cookstove is introduced as energy efficiency measure in the project, therefore equations 1 and 2 of the methodology will be applied to calculate the net GHG emission reductions.

$$ER_y = \sum_i \sum_j ER_{y,i,j} \quad \text{Equation (1)}^{16}$$

Where:

- i = Indices for the situation where more than one type/model of improved cookstove is introduced to replace three-stone fire.
- j = Indices for the situation where there is more than one batch of improved cookstove of type i .
- ER_y = Emission reductions during year y in t CO₂e.
- $ER_{y,i,j}$ = Emission reductions by improved cookstove of type i and batch j during year y in t CO₂e.

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95$$

Where: Equation (2)¹⁷

- $B_{y,savings,i,j}$ = Quantity of woody biomass that is saved in tonnes per improved cookstove of type i and batch j during year y .
- $f_{NRB,y}$ = Fraction of woody biomass that can be established as non-renewable biomass (fNRB)¹⁸.
- $NCV_{wood\ fuel}$ = Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)¹⁹.
- EF_{wf,CO_2} = CO₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO₂/TJ)²⁰.

¹⁶ As Per Equation (1) of applied methodology VMRO006 Version 1.1

¹⁷ As per Equation (2) of applied methodology VMRO006 Version 1.1

¹⁸ C4 Eco Solutions (Pty) Ltd was appointed as third party to study and derive the fNRB value for country specific (Laos) and the report submitted to DOE.

¹⁹ 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction

²⁰ 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion

$EF_{wf,non\ CO2}$ = Non-CO2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO2/TJ)²¹.

$N_{y,i,j}$ = Number of improved cookstoves of type i and batch j operating during year y .

0.95 = Discount factor to account for leakage.

The quantity of woody biomass saved due to implementation of improved cookstoves to be estimated using equation below:

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,y,i,j}}{\eta_{old}} - 1 \right) \quad \text{Equation (3)}^{22}$$

Where:

η_{old} = Efficiency of baseline cookstove

$\eta_{new,y,i,j}$ = Efficiency of the improved cookstove type i and batch j determined through equation 5 of the applied methodology.

$B_{y=1,new,i,j,survey}$ = Annual quantity of woody biomass used by improved cookstoves in tonnes per device of type i and batch j , determined in the first year of the implementation of the project through a sample survey.

$$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94 \quad \text{Equation (4)}^{23}$$

Where:

η_p = Efficiency of project stove (fraction) at the start of project activity.

$(DF_n)^{y-1}$ = Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered.

0.94 = Adjustment factor to account for uncertainty related to project cookstove efficiency test.

Fraction of non-renewable biomass (f_{NRB}) calculation for Laos.

²¹ 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion

²² As per Equation (4) of the applied methodology

²³ As per Equation (5) of the applied methodology

f_{NRB} value derived is 0.6341 which is determined as per TOOL30 “Calculation of the fraction of non –renewable biomass”, version 04. The analysis was independently completed by C4 Ecosolutions (Pty) Ltd, below:

To calculate f_{NRB} we are following the equations as follows

$$f_{NRB} = \frac{NRB}{NRB + RB}$$

$$NRB = H - RB$$

$$H = HW \times N + C + NE$$

$$RB = \sum (MAI_{forest,i} \times (F_{forest,i} - P_{forest,i})) + \sum (MAI_{other,i} \times (F_{other,i} - P_{other,i}))$$

NRB = non-renewable woody biomass

RB = renewable woody biomass production

H = Total annual wood consumption (energy/non energy) in the country

HW = Average HH wood fuel (including charcoal) consumption²⁴

IN = Number of HHs consuming wood fuel in the region/project area.

CE = Commercial woody biomass consumption for energy application.²⁵

NE = Commercial woody biomass consumption for non- energy application

MAI = Mean annual increment in forest/other wooded land²⁶

F = Extent of forest/other wooded land²⁷

P = Extent of protected area (including non-accessible area) within forest/other wooded land.²⁸

²⁴ Approved standardized baseline;Published literature/research paper –Government data, organizations of repute (FAO, UN etc.);Survey; Default value of 0.5 tons/capita/year

²⁵ Energy Statistics Database -UN Statistics Division; Existing studies; Government data; Survey.

²⁶ IPCC Good Practice Guidelines, Table 4.9, www.ipcc-nggip.iges.or.jp

www.fao.org National studies or government data or official statistics.

²⁷ Global Forest Resources Assessment by the Food and Agriculture Organization of the United Nations (FAO); Official statistics: Project specific survey data.

²⁸ Global Forest Resources Assessment by the Food and Agriculture Organization of the United Nations (FAO); National studies or Government data or Official statistics

(i) Estimating Lao's Total woody biomass consumption(H).

The total woody biomass consumption in a country comprises its domestic and non-domestic energy and non-domestic non-energy consumption. For Lao's, this total is estimated using (FAO consumption data) and World Bank population statistics.

The reported volumes of fuelwood consumed in cubic meters are converted to metric tonnes using the FAO default conversion factor. Finally, consumed volumes of commercial forest products—including manufacturing, construction and non-fuel industry are converted to metric tonnes using a biomass conversion and expansion factor provided for the region.

Table 7 : Total population, woody biomass consumption, and per capita consumption estimates for Lao PDR.

Variable	Value	Source ²⁹
Total Population	7,529,475	Urban population - Lao PDR Data (worldbank.org), Rural population - Lao PDR Data (worldbank.org)
Domestic fuelwood consumption(t/yr)	8,903,810	Calculated
Non-domestic (energy) wood consumption(t/yr)	856,188	Calculated
Non-domestic (non-energy) wood consumption (FAO)(t/yr)	3,208,546	Calculated
Conversion factors		
Wood density	0.725	http://www.fao.org/3/a-i6935e.pdf
Biomass conversion and expansion factor	2.050	<u>Global Forest Resources Assessment (FRA) 2020 Lao People's Democratic Republic - Report (fao.org)</u>
Charcoal to wood biomass factor (urban using more efficient kilns)	4	https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-33-v2.0.pdf
H: Total woody biomass consumption (t/yr)	12,968,543	Calculated

(ii) Estimating renewable biomass (RB)

Geospatial data products for Lao were analysed in to estimate the country's renewable biomass. The default age-weighted mean annual increment (MAI) estimates of each ecological zone, as reported by the IPCC, were used for this calculation.

²⁹ f_{NRB} calculation sheet provided to VVB

Table 8:. Total, protected and remote forest cover extent, mean annual increment (MAI), and renewable biomass (RB) by Global Ecological Zone for Lao PDR.

FAO Global Ecological Zone	Total forest cover(ha)	Protected area cover(ha)	Remote area cover(ha)	MAI (t/ha/yr)	Annual growth(t/yr)
Tropical dry forest	1,539,236	45,942	732,589	1.19	906,038
Tropical moist forest	3,900,208	15,700	2,091,418	0.70	1,254,464
Tropical mountain system	3,009,403	380,096	1,351,394	0.17	218,048
Tropical rainforest	4,415,536	387,761	2,287,791	1.36	2,366,998
Total	12,864,383	829,498	6,463,192	–	4,745,548

(iii) The fraction of non-renewable biomass (f_{NRB})

The fraction of non-renewable biomass is the quotient of the non-renewable and the total biomass. The fraction of non-renewable biomass for Lao is, therefore, estimated to be 0.6341

Table 9:. Fraction of non-renewable biomass (f_{NRB}) calculation for Lao PDR.

Variable	Unit	Value	Source
Total woody biomass consumption (H)	t/yr	12,968,543	Table 7 above
Renewable biomass (RB)	t/yr	4,745,548	Table 8 above
Non-renewable biomass (NRB) i.e., $NRB = (H - RB)$	t/yr	8,222,996	Calculated
Fraction of non-renewable biomass (f_{NRB}) i.e., $f_{NRB} = (NRB / (NRB + RB))$	-	0.6341	Calculated

(iv) Item 6(b)(i) of tool 30 Version 04 Comparison of peer literature and derived f_{NRB}

The resulting f_{NRB} value is higher than the expected f_{NRB} values determined in peer reviewed literature such as “The carbon footprint of traditional woodfuels” by Bailis and colleagues using the WISDOM method. While direct comparisons between the WISDOM and CDM methodologies are only sometimes appropriate, given the different approaches of the methodologies, the following factors can contribute to variations in the f_{NRB} estimates between the application of the two methods:

1. More recent population statistics (2022) for Lao PDR were used in the present study. The more recent data represents an increase in population numbers since the 2009 data utilized in the study by Bailis and colleagues 2015.
2. Updated 2020 FAO forest products statistics, UNSTAT data were used in the present study, whereas the study conducted by Bailis and colleagues used 2013 FAO forest products statistics.

3. The approach used to determine the amount of forest that is accessible yields lower estimates in the present study when compared to the study by Bailis and colleagues. This results in a lower estimated RB and consequently increased NRB and f_{NRB} .

4. In the present study, MAIs were calculated using a weighted average based on the forest area of three categories (i.e., primary forests, above 20-year secondary forest, below 20-year secondary forest). Data from the 2019 Refinement of the 2006 IPCC Guidelines was used in combination with extrapolating the observed forest gain extents between 2000 and 2012 to a future 20-year period. As per the study by Bailis and colleagues, MAI values were derived from a combination of field observations and IPCC values, followed by a different estimation of growth rates as a percentage of standing stock. This approach often yields higher MAIs and may lead to higher estimations of RB and subsequently, lower estimations of NRB and f_{NRB} .

(v) Item 13 of tool 30 Version 04 Crosscheck of NRB ³⁰

The resulting non-renewable biomass (NRB) value was compared with the top-down product of average above-ground biomass tonnage of the forest are deforested (121.52 t/ha)³¹ and the latest average annual deforestation (322,000 ha/yr.)³² which results in the NRB value 39,123,000 t/yr as per deforestation rate. NRB, as calculated for Laos PDR according to the latest CDM Tool 30 Version 04³³ is 8,222,996 t/yr. The NRB estimated is less (-376%) than value as per the product of biomass and deforestation rate. Thus NRB estimate is deemed relevant for this study, as it falls within a 10% margin of difference from the cross-check, in accordance with the guidelines outlined in CDM Tool 30 v4.0 2022.

Project Activity Instance

For the present project activity instance emission reduction is calculated using above equations:

Parameter	Description	SSM-S32-X	Naga Cookstove	Unit	Source
$f_{NRB, y}$	Fraction of woody biomass saved by the project activity in year y that can be established as non-renewable biomass.	0.6341	0.6341	Fraction	Calculated based on CDM Tool 30 version 4
NCV _{biomass}	Net Calorific Value of biomass.	0.0156	0.0156	TJ/tonne	Methodology default

³⁰ f_{NRB} calculation sheet provide to VVB

³¹ Page 41 : [Global Forest Resources Assessment \(FRA\) 2020 Lao People's Democratic Republic - Report \(fao.org\) \(item 2c pg46\)](#)

³² For year 2022 [Laos Deforestation Rates & Statistics | GFW \(globalforestwatch.org\)](#)

³³ NRB value as given in table 9 of this VCS PD

EF_{CO_2}	CO2 emission factor for wood fuel.	112	112	tCO ₂ e/TJ	Methodology default
EF_{non-CO_2}	Non-CO2 emission factor for wood fuel.	26.23	26.23	tCO ₂ e/TJ	Methodology default
L	Leakage adjustment factor.	0.95	0.95	Fraction	Methodology default
η_{old}	Efficiency of the system being replaced.	0.10	0.10	Fraction	Methodology default
D_{Fn}	Discount factor.	0.99	0.99	Fraction	Methodology default
η_p	Thermal efficiency of the project cookstove at the start of the project activity.	0.5150	0.3500	Fraction	Manufacturer's specification
$\eta_{new,i,y}$	Thermal efficiency of the project cookstove during the first year of the project crediting period (01 July 2023- 30 June 2024).	0.4841	0.3290	Fraction	calculated using equation (5) mentioned in VMR0006 version 1.1
$N_{y,i,a}$	Number of stoves still in operation during the monitoring period.	250,000	250,000	Quantity	assumed
$B_{y=1, new,i,survey}$	Annual quantity of woody biomass used during the project activity in tonnes per cookstove.	1.0841	0.6852	Tons/year	Assumed based on similar project in Laos (VCS ID: 2521

For ex-ante calculation purpose, the assumption below is applied.

- 1) The project will install up to 250,000 SSM-S32-X and 250,000 Naga Cookstoves.
- 2) Annual Stove Loss Rate is estimated at 5%.

- 3) $B_{y=1,new,i,j,survey}$ is assumed as 1.0841 tonnes/device/year (for SSM S-32X) and 0.6852 tonnes/device/year (for Naga Cookstove) from previous projects (VCS 2521- MP 02).

Determination of number of ICS operating during year y

$$N_{y,i,j} = N_{y-1,i,j} \times [1 - 5\%]$$

Example of calculation:

If $y = 5$,

$$N_{y,i,j} = 232,510 \times [1 \times 5\%]$$

$$= 232,510 \times 95\%$$

$$= 232,510$$

Hence, the number of ICS operating during year y is as below:

Year	SSM-S32-X	Naga Cookstove
1	48,400	48,400
2	117,980	117,980
3	183,428	183,428
4	232,510	232,510
5	220,884	220,884
6	209,840	209,840
7	199,348	199,348
8	189,380	189,380
9	179,911	179,911
10	170,916	170,916

Calculation of annual drop in efficiency

Parameter	Description	Value	Unit	Source
η_p	Thermal efficiency of the project cookstove	0.5150	Fraction	Manufacturer's specification
DF_n	Drop in efficiency	0.99	Factor	Methodology default

y	Current year for which efficiency is to be determined.	-	year	-
---	--	---	------	---

SSM-S32-X Stove Efficiency Loss and Emission Reduction Projection

Emission Reduction Projection				
Years	$\eta_{new,i} = \eta_p \times (DF_n)^{y-1} \times 0.94$	Efficiency ($\eta_{new,y}$)	B _{y, savings}	ER _y
1	$\eta_{year\ 1} = 0.5150 \times 0.99^{1-1} \times 0.94$	48.41%	4.1640	261,788
2	$\eta_{year\ 2} = 0.5150 \times 0.99^{2-1} \times 0.94$	47.93%	4.1115	630,092
3	$\eta_{year\ 3} = 0.5150 \times 0.99^{3-1} \times 0.94$	47.45%	4.0596	967,249
4	$\eta_{year\ 4} = 0.5150 \times 0.99^{4-1} \times 0.94$	46.97%	4.0082	1,210,527
5	$\eta_{year\ 5} = 0.5150 \times 0.99^{5-1} \times 0.94$	46.50%	3.9572	1,135,390
6	$\eta_{year\ 6} = 0.5150 \times 0.99^{6-1} \times 0.94$	46.04%	3.9068	1,064,880
7	$\eta_{year\ 7} = 0.5150 \times 0.99^{7-1} \times 0.94$	45.58%	3.8569	998,712
8	$\eta_{year\ 8} = 0.5150 \times 0.99^{8-1} \times 0.94$	45.12%	3.8075	936,622
9	$\eta_{year\ 9} = 0.5150 \times 0.99^{9-1} \times 0.94$	44.67%	3.7586	878,359
10	$\eta_{year\ 10} = 0.5150 \times 0.99^{10-1} \times 0.94$	44.22%	3.7102	823,690

Naga Stove Efficiency Loss and Emission Reduction Projection

Emission Reduction Projection				
Years	$\eta_{new,i} = \eta_p \times (DF_n)^{y-1} \times 0.94$	Efficiency ($\eta_{new,y}$)	B _{y, savings}	ER _y
1	$\eta_{year\ 1} = 0.350 \times 0.99^{1-1} \times 0.94$	32.90%	1.5691	98,648
2	$\eta_{year\ 2} = 0.350 \times 0.99^{2-1} \times 0.94$	32.57%	1.5466	237,010
3	$\eta_{year\ 3} = 0.350 \times 0.99^{3-1} \times 0.94$	32.25%	1.5242	363,171
4	$\eta_{year\ 4} = 0.350 \times 0.99^{4-1} \times 0.94$	31.92%	1.5022	453,674
5	$\eta_{year\ 5} = 0.350 \times 0.99^{5-1} \times 0.94$	31.60%	1.4803	424,715

6	$\eta_{year\ 6} = 0.350 \times 0.99^{6-1} \times 0.94$	31.29%	1.4586	397,577
7	$\eta_{year\ 7} = 0.350 \times 0.99^{7-1} \times 0.94$	30.97%	1.4372	372,147
8	$\eta_{year\ 8} = 0.350 \times 0.99^{8-1} \times 0.94$	30.66%	1.4160	348,318
9	$\eta_{year\ 9} = 0.350 \times 0.99^{9-1} \times 0.94$	30.36%	1.3950	325,992
10	$\eta_{year\ 10} = 0.350 \times 0.99^{10-1} \times 0.94$	30.05%	1.3741	305,074

Year	Estimated baseline emissions or removals (tCO _{2e})	Estimated project emissions or removals (tCO _{2e})	Estimated leakage emissions (tCO _{2e})	Estimated net GHG emission reductions or removals (tCO _{2e})
17-February-2024 to 31-December-2024	-	-	-	315,011
01-January-2025 to 31-December-2025	-	-	-	803,247
01-January-2026 to 31-December-2026	-	-	-	1,272,029
01-January-2027 to 31-December-2027	-	-	-	1,622,135
01-January-2028 to 31-December-2028	-	-	-	1,573,224
01-January-2029 to 31-December-2029	-	-	-	1,474,762
01-January-2030 to 31-December-2030	-	-	-	1,382,402
01-January-2031 to 31-December-2031	-	-	-	1,295,768
01-January-2032 to 31-December-2032	-	-	-	1,214,507
01-January-2033 to 31-December-2033	-	-	-	1,138,290
01-January-2034 to 16-February-2034	-	-	-	142,255
Total	-	-	-	12,233,630

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$f_{NRB,y}$
Data unit	Fraction
Description	Fraction of woody biomass saved by the project activity during year y that can be established as non-renewable biomass.
Source of data	Value calculated using CDM Tool 30; version 04
Value applied	0.6341
Justification of choice of data or description of measurement methods and procedures applied	This parameter is determined ex-ante. C4 EcoSolutions was appointed as third party to study and derive the f_{NRB} value for Lao PDR.
Purpose of Data	Calculation of emission reductions.
Comments	-

Data / Parameter	$NCV_{wood\ fuel}$
Data unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass that is substituted or reduced.
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction.
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value.
Purpose of Data	Calculation of emission reductions.
Comments	-

Data / Parameter	$EF_{wf,CO2}$
Data unit	tCO ₂ /TJ
Description	CO ₂ emission factor for the use of wood fuel in baseline scenario.
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion.
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value.
Purpose of Data	Calculation of emission reductions.
Comments	-

Data / Parameter	$EF_{wf,non\ CO2}$
Data unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor for the use of wood fuel in baseline scenario.
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion.
Value applied	26.23
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	-

Data / Parameter	η_p
Data unit	Fraction
Description	Efficiency of project stove(fraction) at the start of project activity.

Source of data	Manufacturer's specification.
Value applied	For SSM-S32-X Cookstove: 0.5150 For Naga Cookstove: 0.350
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante as per the manufacturer's specification.
Purpose of Data	Calculation of $\eta_{new, y, i, j}$
Comments	-

Data / Parameter	D_{Fn}
Data unit	Fraction
Description	Discount factor to account for efficiency loss of project cookstove.
Source of data	Default Value
Value applied	0.99
Justification of choice of data or description of measurement methods and procedures applied	Default value as per applied methodology.
Purpose of Data	To account for efficiency loss of project cookstove per year of operation.
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$N_{y, i, j}$
Data unit	Number
Description	Number of project devices of type I and batch j operating during year y.
Source of data	Monitoring
Description of measurement methods	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve

and procedures to be applied	90/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and Programme of activities.
Frequency of monitoring/recording	At least once every two years.
Value applied	For ex-ante emission reduction calculation, it is assumed that the project will distribute up to 250,000 SSM-S32-X stoves and 250,000 Naga Stoves in present project activity instance. For the ex-ante estimation of ERs, PP has considered 5% non-operational stoves and deducted each year conservatively.
Monitoring equipment	Not applicable.
QA/QC procedures to be applied	-
Purpose of data	Calculation of emission reductions.
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value.
Comments	-

Data / Parameter	η_{old}
Data unit	Fraction
Description	Efficiency of baseline cookstove
Source of data	Methodological default value
Description of measurement methods and procedures to be applied	As per the applied methodology default value of 0.1 shall be used if baseline device is a three-stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney.
Frequency of monitoring/recording	Fixed for each individual household at the time of project implementation.
Value applied	0.1
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of emission reductions.

Calculation method	none
Comments	

Data / Parameter	$\eta_{new,y,i,j}$		
Data unit	Fraction		
Description	Efficiency of the device of each type i and batch j implemented as part of the project activity.		
Source of data	Calculation		
Description of measurement methods and procedures to be applied	To adopt Option V given in the methodology: “Efficiency of the improved cookstoves to be estimated using equation 5 of the applied methodology where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored”		
Frequency of monitoring/recording	Annually		
Value applied	For ex-ante calculation, the value below is applied.		
	Year (y)	SSM-S32-X	Naga Cook Stove
	1	48.41%	32.90%
	2	47.93%	32.57%
	3	47.45%	32.25%
	4	46.97%	31.92%
	5	46.50%	31.60%
	6	46.04%	31.29%
	7	45.58%	30.97%
	8	45.12%	30.66%
	9	44.67%	30.36%
	10	44.22%	30.05%
Monitoring equipment	Not Applicable.		
QA/QC procedures to be applied	Not Applicable.		

Purpose of data	Calculation of emission reductions.
Calculation method	Calculation to be performed using equation below: $\eta_{\text{new},y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$
Comments	-

Data / Parameter	$B_{y=1,new,i,j,survey}$
Data unit	Tonnes
Description	Quantity of woody biomass used by project devices tonnes per device of type i
Source of data	Monitoring survey
Description of measurement methods and procedures to be applied	Minimum sample size of each type i and batch j should be in line with the latest version of Standard for sampling and surveys for CDM project activities and Programme of activities or guidelines provided in methodology Section 8.4 option (b). Determined in the first year of the introduction of the devices (e.g., during the first year of the crediting period, $y=1$) through measurement campaigns at representative households and/or sample survey. Sample surveys to estimate this parameter, that are solely based on questionnaires or interviews (i.e., that do not implement measurement campaigns) may only be used if the following conditions are satisfied. (i) Baseline cookstoves have been completely decommissioned and only improved cookstoves are exclusively used in the project households. (ii) If multiple devices are used in the project, it is possible from the results of the survey questions to clearly differentiate the quantity of firewood being used by each device. In other words, if more than one device, or another device that consumes firewood, are in use in project households, then the sample survey needs to distinguish the quantity of firewood used by the project device and the other devices that use firewood.
Frequency of monitoring/recording	Determined in the first year of project implementation.
Value applied	For ex-ante calculation, the value for SSM-S32-X Stove is assumed as 1.0841 Tonnes/device/year and for Naga Cookstove is assumed as 0.6852 Tonnes/device/year.

Monitoring equipment	Measurements undertaken using weighing Scales and moisture meters for quantity of fuel wood. Calibration of weighing scales and moisture meters to be used for measuring the fuel wood will get done before start using on site.
QA/QC procedures to be applied	Survey to be conducted as per requirements of the methodology. Monitoring equipment will be calibrated as per the manufacturer's guidelines.
Purpose of data	Calculation of emission reductions.
Calculation method	-
Comments	-

Data / Parameter	Life Span
Data unit	Number of years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency.
Source of data	Manufacturer's specification
Description of measurement methods and procedures to be applied	Not required
Frequency of monitoring/recording	Once at the time of project stove installation.
Value applied	10 years for SSM-S32-X and more than 3 years for Naga Cookstove.
Monitoring equipment	Not applicable
QA/QC procedures to be applied	Not applicable
Purpose of data	-
Calculation method	Not applicable

Comments	Calculation of Emission Reductions
Data / Parameter	Date of commissioning of project device i.
Data unit	Date
Description	Actual date of commissioning of the project device
Source of data	Project database
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	Fixed and recorded at the time of commissioning/distribution
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of emission reductions
Calculation method	-
Comments	none

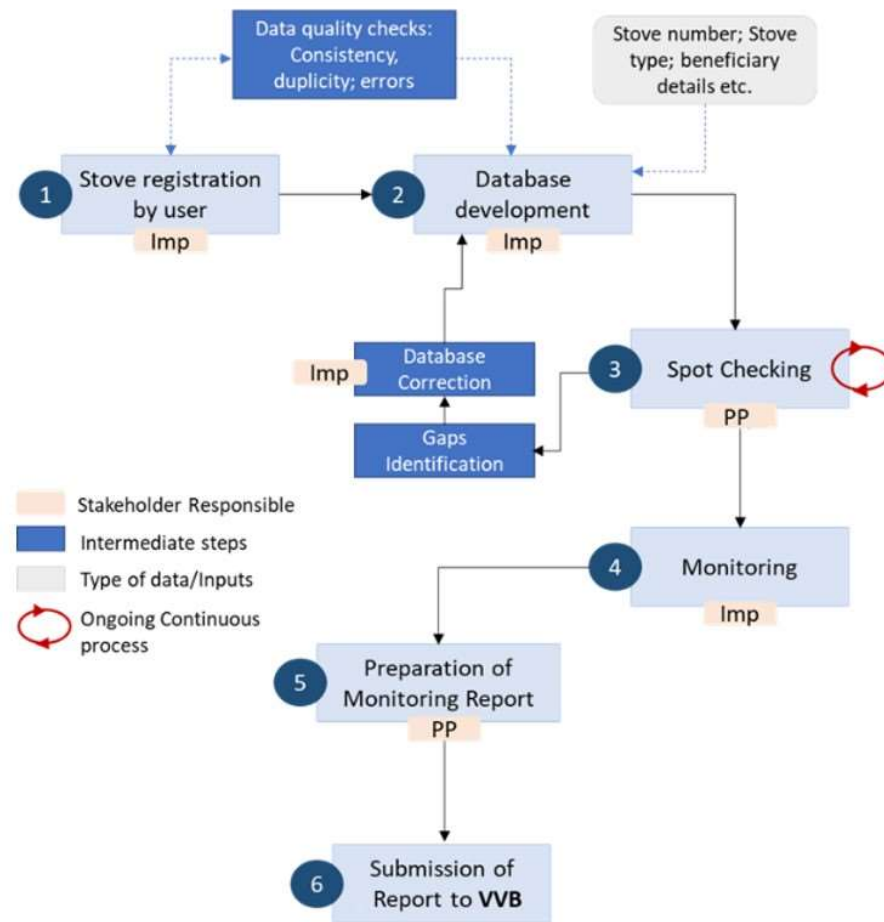
5.3 Monitoring Plan

The Monitoring Plan applied in this project involves several key elements that ensure that the PP and/or project Implementer have high-quality, unbiased, and reliable information regarding the performance of the project in terms of implementation and outcomes, and for the purposes of calculating Verified Carbon Units (VCUs) following VCS methodology (VMR0006 version 1.1) on the basis of the amount of non-renewable biomass saved by the ICS in the project activity. The key elements are the following:

- Data collection procedures
- Distribution and Monitoring Database
- Spot Checking of ICS (ongoing)
- Sample Plan for the Monitoring Survey
- Data Quality, Consistency and Duplication Checks

- Monitoring Reporting

The below flow-chart illustrates the roles and responsibilities of the parties during the implementation of the monitoring plan for the project. In the below flowchart, the project implementer is abbreviated to “Imp” and can be the PP or another party authorized by the PP.



Roles and Responsibilities of stakeholders during the implementation of the monitoring plan

Below is the description of the above steps on the flow-chart.

- Implementor: User registers stove:** Project implementer will collect/receive the necessary information requested in the Registration process from the user. Means of collecting this information may be through a physical registration card filled by project Implementor staff, retailers, end-users, or partner organization’s staff, or using ICTs or SMS. Project Implementers’ staff shall double check the accuracy of information provided, and request for field staff additional clarifications if needed.
- Implementor: Data logged into database:** Project implementer’s trained staff will input the data in the database either manually (if data collected from physical registration card) or this will be automatically input if data was collected using ICTs or SMS. Project implementer staff shall double check the information included on the database and check for duplications. Any

duplicate information shall be investigated, and errors corrected or excluded from the database if it is a true duplicate entry.

3. **Project Proponent: Spot- checking (ongoing):** Project Proponent has a procedure for internal auditing called spot-checking, where field staff randomly selects households to ensure proper functioning of the project stoves throughout the project lifetime and to ensure that the project stoves continue to work as efficient as specified by the manufacturer. Field staff of project proponent randomly select units included in the database and visit or contact the stove owners to cross-check the information on the database with the factual evidence in the field. Any inconsistencies found (e.g., change in the address of a user) gets updated on the database, and in case, ICS are found to be no longer in use, they get clearly marked as such and is excluded from emission reductions calculations.

During the spot check, if any of the project stoves was found to be in poor working condition like damaged or in unusable condition, the stove owner may repair the stove for minor issues. If the stove owners are unsuccessful in repairing or solving the issues on their own, the owner may reach out to CQC & IP through helpline, field personnel or the village headmen for further assistance. Respective channel representative, either headman, field staff, IP, helpline executive or authorized personnel informs CQC or their officials for appropriate action and to resolve the grievance raised by the beneficiary. CQC or their representatives will keep a record of such incidents and ensure the resolution is achieved to the satisfaction of the end users.

4. **Implementor: Monitoring:** Project implementer will follow the requirements as per PD to collect the necessary information for a monitoring report. The project proponent will assess all monitoring data and produce a monitoring report for the VVB to verify corresponding to the preceding monitoring period. This report will present the data relating to the emission reductions generated by those batches during the monitoring period.
5. **Project Proponent: Preparation of monitoring report:** the project implementers or the PP will prepare the final monitoring report to be provided to the VVB for verification of emission reductions. A copy of the monitoring report will remain with the Project Proponent.

The PP will coordinate and manage each project Implementer and assist them in implementing each element of the monitoring plan. The monitoring plan shall be elaborated in accordance with the Sampling Plan below. Also, regular audits conducted throughout the project lifetime ensures proper functioning of stoves and determination of the lifespan of the same.

Sampling Plan

As per the Guideline for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 04.0, the sampling plan is the following:

A. Sampling design

Due to the large number of ICS envisioned to be distributed under the project, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling shall be undertaken for each batch of ICS. The sampling plan to be designed in line with the requirements of the Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 09.0.

Samples will be drawn from ICS recorded in the project database administered by the project proponent. A detailed explanation of this database is found in Section 1.11 of project description.

i. Objectives and Reliability Requirements

The objective is to obtain an unbiased and reliable estimate of the proportion or mean value of the following key variables over the course of the crediting period.

Monitored Parameters:

The project involves introduction of improved cookstoves as energy efficiency measure.

No.	Monitoring parameters	Sampling parameters	Parameter type	Monitoring frequency
1	$N_{y,i,j}$ Number of project devices of type i and batch j operating during year y	Proportion of ICS still in operation	Proportion	At least once in two years.
2	$B_{y=1,new,i,j,survey}$ Quantity of woody biomass used by project devices in tonnes per device of type i . and batch j	Daily consumption of woody biomass per ICS	Mean value	Determined in the first year of project implementation

ii. **Target Population**

The target population will be the complete set of appliances (ICS) deployed under the project.

iii. **Sampling Frame**

It is expected that the geographical locations do not have influence on the parameters of interest that is $N_{y,i,j}$ & $B_{y=1,new,i,j,survey}$. Therefore, these parameters can be assumed to be highly homogeneous regardless of how the end user group and distribution/installation location is defined since the end users shall always be households and geographical location is restricted to the project boundary. The Sampling Frame for individual parameters have been defined in the following paragraphs:

The sample frame refers to all the information sources on the Database. The SMS/ICT system is the primary mechanism for data collection for newly distributed ICS and the Monitoring Survey (which includes a household questionnaire and visual inspection of ICSs) that will be used throughout the lifetime of the project. The SMS/ICT data is used to populate the stoves Database and the Monitoring Survey follows the “Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities”; version 09.0.

All the ICS distributed under this project shall be grouped together to create a Primary Sampling Unit which is homogenous. The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error. Using, a 90 per cent confidence level, and a 10 per cent margin of error, a random sample will be selected from the target population.

a. **Sampling frame for proportion of ICS still in operation ($N_{y,i,j}$)**

The first step is to identify the Primary Sampling Units.

Factors such as model of the ICS do not effect parameter $N_{y,i,j}$. Therefore Primary sampling unit in case of parameters $N_{y,i,j}$ is independent of model of stove.

b. Quantity of woody biomass used by project devices in tonnes per device ($B_{y=1,new,i,survey}$)

The Parameter $B_{y=1,new,i,survey}$ shall vary in accordance with its model. Hence for this parameter, the Primary Sampling Unit shall be defined as the group of ICSs of the same model. If two different ICS models being implemented in the same project – this will form two Primary Sampling Units. The value will be determined in the first year of the introduction of the devices through the monitoring survey.

Under this project upto two ICS will be installed in each household, which are classified as Project stove 1 and project stove 2. At the time of monitoring survey, field staff record the same in survey app. Therefore, firewood consumed for each project stove can be distinguished clearly. For measuring the fuel wood, calibrated weighing scales will be used to ensure data quality and accuracy.

iv. Sampling Method

The sampling method for the monitored parameters $N_{y,i,j}$, and $B_{y=1,new,i,survey}$ is Simple Random Sampling and samples will be

randomly selected from the primary sampling units as illustrated above. To ensure a random selection of ICS, random number generators shall be applied. Each ICS in the target population is uniquely identifiable by its unique ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined sampling frame. Applying the random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size as calculated by the project proponent.

To determine the parameters, sampling will involve the following approaches (outcome in brackets):

$N_{y,i,j}$	Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use (Yes/No)
$B_{y=1,new,i,j,survey}$	Quantity of woody biomass used by project devices in tones per device (First year of installation).

v. Sample Size

For the estimation of the proportion or mean value of the parameters investigated, the minimum sample size for each sample frame must achieve the 90/10 confidence/precision.

The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error. Using a 90 per cent confidence level, and a 10 per cent margin of error, random samples will be selected from each Primary Sampling Unit. The parameter number of stoves still in operation during the monitoring period as determined by the monitoring survey ($N_{y,j}$) will be estimated through sampling annually/biennially and the parameter Quantity of woody biomass used by project devices in tonnes per device $B_{y=1,new,i,survey}$ shall be estimated through sampling within the first year of project installation. $B_{y=1,new,i,survey}$ can

be coupled with $N_{y,j,j}$ for the first year. Of the two parameters to be monitored, $N_{y,i,j}$ is proportion/percentage parameter and $B_{y=1,new,i, survey}$ is a mean value parameter.

To calculate the required sample size estimates, values for the proportions, mean values, and standard deviations are required. As per Guideline for Sampling and surveys for CDM project activities and programmes of activities, version 04.0, there are different ways available to obtain the estimates of the parameter of interest:

- (a) Refer to the result of previous studies and use these results;
- (b) In a situation where information from previous studies is not available, a preliminary sample as a pilot could be conducted and use that sample is used to provide the estimates;
- (c) Use best guesses based on the researcher's own experiences.

For the registration purpose of project, option (c) as stated above shall be applied. For the following monitoring periods, the estimates can be adjusted taken the results of the previous monitoring period(s) into account or the result from recent pilot study which is conducted after the previous monitoring periods.

Parameter $N_{y,i,j}$:

To estimate the sample size for the parameter $N_{y,i,j}$ the following equation is used:

$$n \geq \frac{1.645^2 N \times p \times (1-p)}{(N-1) \times 0.1^2 \times p^2 + 1.645^2 p \times (1-p)}$$

Where:

- | | |
|-------|--|
| n | = Sample size |
| N | = Population size (Total number of households/ICS) |
| p | = Expected proportion |
| 1.645 | = Represents the 90% confidence required |
| 0.1 | = Represents the 10% relative precision |

The following assumptions are made to exemplify the sample size calculation for the parameters.

1. An overview of the estimated sample sizes for a hypothetical population of 100,000 ICS units applying a level of 90/10 is provided below. It is likely that all the sample frames for each parameter will include fewer than 100,000 ICS in the first monitoring period, so this is a conservative approach. Hence, population size, N, is taken as 100,000 households/ICS (Assuming one ICS for one household).
2. It is expected at least 80% of ICS still in operation, hence the expected proportion p is taken as 0.8.

Sample size calculation:

The calculation of the required sample size for each parameter in the first monitoring period is illustrated below. In all cases a conservative approach is taken, however if for any parameter the required confidence/precision is not met then the PP will randomly select an additional sample and collect further data from this sample to ensure the pooled data meet or exceed the required thresholds.

Based on the above assumptions, the resulting sampling size is calculated as:

$$n \geq \frac{1.645^2 \times 100,000 \times 0.8(1 - 0.8)}{(100,000 - 1) \times 0.1^2 \times 0.8^2 + 1.645^2 \times 0.8(1 - 0.8)} = 67.61$$

Therefore, in this case a sample size of 68 is to be sampled from primary sampling unit.

Parameter $B_{y=1, \text{new}, i, \text{survey}}$:

For the purposes of determining sample size in the first monitoring period, the performance of ICS is characterized by the range of likely mean fuel wood consumption and the likely values of SD relative to the mean, according to the type of ICS.

To estimate the sample size for parameter $B_{y=1, \text{new}, i, \text{survey}}$ the following equation is used:

$$n \geq \frac{1.645^2 NV}{(N-1) \times 0.1^2 + 1.645^2 V}$$

Where:

$$V = \left(\frac{SD}{mean} \right)^2$$

n	= Sample size
N	= Population size (Total number of households/ICS)
mean	= Expected mean of fuel wood consumption
SD	= Expected standard deviation
1.645	= Represents the 90% confidence required
0.1	= Represents the 10% relative precision

Based on the above assumptions, the sample size calculation would be

$$n \geq \frac{1.645^2 \times 100,000 \times \left(\frac{0.076}{0.38} \right)^2}{(100,000 - 1) \times 0.1^2 + 1.645^2 \times \left(\frac{0.076}{0.38} \right)^2} = 10.82$$

The resulting sample size based on the above equation is smaller than 30, since $B_{y=1, \text{new}, i, \text{survey}}$ is a numeric mean value (i.e. not a proportion or percentage) the Student's t-distribution shall be

used as per paragraph 14 of “Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 09.0.

The sample size for parameter $B_{y=1, \text{new}, i, \text{survey}}$ under t-distribution is referred to the equation below:

$$n = \left(\frac{t_{n-1} \times SD}{0.1 \times \text{mean}} \right)^2$$

Where t_{n-1} is the value of the t-distribution for 90% confidence when the sample size is n. Since the sample size is not known yet, the first step is to use the value for 90% confidence when the sample is large, i.e., 1.645 and then redefine the calculation.

$$n = \left(\frac{1.645 \times 0.076}{0.1 \times 0.38} \right)^2 = 10.82$$

Thus, n is rounded up to 11.

The calculation now needs to repeat using t-value for 90% confidence and n = 11

$$n = \left(\frac{2.228 \times 0.076}{0.1 \times 0.38} \right)^2 = 19.86$$

And n is rounded to 20.

The calculation now needs to repeat using t_{n-1} value for n = 20. The process should be iterated until there is no change to the value of n.

T_{20-1}	2.093
n=	17.52
Round up	18

t_{18-1}	2.110
n=	17.81
Round up	18

The repeated calculation shows that n = 18. Thus, the sample size to be sampled from each sampling unit is 18.

Since parameters $N_{y,j,j}$, and $B_{y=1, \text{new}, i, \text{survey}}$ share the same sampling units, PP may choose to have one common survey for these two parameters with largest number of sample size between these two parameters being chosen. Sampling more than one parameter within the same sample (household) helps reduce travel needs for monitoring and the associated costs. At the same time this approach ensures the random selection of samples for every parameter.

It is a good practice to employ oversampling at the design stage, not only to compensate for any attrition, outliers or non-response associated with the sample, but also to prevent a situation at the analysis stage where the required reliability is not achieved, and additional sampling efforts would be required. The sample size shown above will be adjusted upwards to account for non-responses, PP shall determine the appropriate non-responses rate based on previous experience.

B. Data:

i. Field Measurements:

To monitor the number of stoves that continue to be in use ($N_{y,i,j}$), the data collected will be a representative number of stoves in the database that are in use for the monitoring period. The method of collecting data will be field surveys of required sample size of ICS users in the database. Data will be collected from the field surveys, entered in the database and included in the monitoring report.

PP keeps records of all distributed ICS in an electronic database. As a minimum the following information will be recorded in the database:

1. Date of distribution/installation
2. Geographic area of distribution/installation (geodetic coordinates of ICS household)
3. Model/type of project technology distributed/installed (ICS model)
4. Quantity of project technologies distributed/installed
5. Name and telephone number (if available), and address of recipient
6. unique identification alpha/numeric ID for each device that is sold/distributed (ICS serial number)

For monitoring $B_{y=1,new,i, survey}$, the data collected will be representative number of stoves in the database that were distributed within one year of start date of project implementation. The method of collecting data will be field survey and frequency will be once during the entire crediting period.

The table below summarizes field measurement data requirements

Parameter	Timing (indicative)	Frequency as per methodology	Methods to be applied	Comments on seasonal fluctuation
$N_{y,i,j}$	Monitoring will likely occur at least once in every 12 months	At least once in two years.	Visits to the end user premises, visual inspection and interview with ICS end-user	Unlikely to be due to any seasonal fluctuation.
$B_{y=1,new,i, survey}$	Within 1 st year from the starting date of	Once. The value will be fixed for	Visits to the premises, visual inspection, measurement and	unlikely

	project implementation	entire crediting period	interview with ICS end- user.	
--	---------------------------	----------------------------	----------------------------------	--

PP conducts an annual survey for sites included in the project to identify the number of cookstoves that remain in operation.

The survey will also obtain, at a minimum, the following information:

1. The cookstoves distributed under the project are being used.
2. The project stoves are operational and in good condition

ii. Quality Assurance/Quality Control

The PP will apply measures to ensure the required confidence/precision for each sampled parameter is met, allowing for non-response and the possible removal of outliers from the sample, as part of a Quality Control/Quality Assurance system. The choice of measure applied to each parameter will depend on the cost of each data collection approach and logistics required. The PP will determine the most effective measure for each parameter from the following list (illustrated using a required sample size of 20 and an effect of non-response of 2 to 4 ICS):

- Oversampling: Randomly draw a sample of minimum 24 ICS and collect data from each
- Buffer Group: Randomly draw a sample of at minimum 24 ICS and collect data from only 22 ICS. If this would not result in the required sample size data would be collected from the additional 2 ICS that were selected in the sample.
- Draw an additional sample: Randomly draw a sample of 22 ICS and collect data from these. If the required sample size is not achieved, an additional sample of 2 elements will be drawn and included in the sample.
- Use lower confidence bound (of $N_{y,i,j}$, $B_{y=1,new,i,survey}$) or, with a conservative approach according to the parameter definitions,

The PP may choose to stop monitoring a particular parameter once the required level of confidence/precision has been reached, as long as the calculated minimum number of samples has been achieved. As an example, the following steps could logically be followed for the case of applying a 30% buffer:

- a. Visit first 10% of premises required for the 30% buffer. If the number of responses is sufficient to achieve the required reliability level, then stop sampling.
- b. If step 1 is not sufficient to achieve the required reliability level, then visit the next 10% of premises (increases the additional sampling to 20% of the 30% buffer). If this additional sampling is sufficient, then stop sampling.
- c. If step 2 is not sufficient to achieve the required reliability level, then complete the final 10% of the additional sampling buffer (bringing the total to 30%).

The sampling plan has the following procedures in place to ensure good quality data. The PP will ensure that field personnel have reviewed, understood, and signed the monitoring plan, including provisions for maximizing response rates, documenting out-of-population cases, refusals and other sources of non-response. A quality control and assurance strategy will be documented. Quality control and assurance strategies include addressing non-sampling errors, such as non-response

or bias from interviewer. The monitoring plan will explain how to properly survey households to prevent bias from interviewer. In the case a household refuses to participate, another household will be chosen at random. To reduce interviewer bias, good questionnaire design and well-tested questionnaires will be used.

The calculation of the sample size will be carried out using estimates for parameter proportions, mean values and standard deviations, as the actual characteristics of the population/sampling frame are unknown. In order to ensure the quality of the sampling results, the PP can draw on the provisions for reliability calculations as provided by the Guidelines for Sampling and Surveys in CDM Project Activities and Programme of Activities (version 04). In the event that the sampling results do not fulfil the required level of confidence and precision, the PP can undertake additional samples. If the reliability is still not sufficient after additional samples or other measures, the sampling may be repeated with an increased sample size. Alternatively, the PP may choose to apply the lower bound or higher bound according to the more conservative approach.

iii. Data archiving

Hard copies of the surveys will be kept and the registration database will have back up. Original stove purchase contracts or other means of acceptance by the users will be stored in the main office for the coordinating entity. A back-up of the registration database will also be stored on an electronic medium by the PP. All data monitored and required for verification and issuance will be kept for two years after the end of the crediting period or the last issuance of VCUs for the project activity, whichever is later.

iv. Analysis

The PP will manage a project database that includes the following data that can be directly attributable to the project activity, thereby allowing unambiguous determination of the emission reductions attributable the project:

- A list of households participating in project activity, including name, community/location, distribution date and unique serial number;
- Where replacements are made, assurance that the efficiency of the new ICS is similar to the specified.

Data obtained from the samples will be used to estimate proportions and mean values for the parameters described above. The values will then be factored into the emissions reduction calculations and result in the request for issuance of VCUs. The stoves that are not in use will be excluded from emissions reductions calculations and will not be counted towards the total number of ICS in operation during the monitoring period.

C. Implementation

Sampling for the purpose of emission reduction calculation and elaboration of the monitoring report will occur at the end of each monitoring period. This sampling will be conducted by trained personnel, either part of the project Implementer or PP team, or an experienced third-party entity. The maximum length of one monitoring period will be two years (duration, not calendar years). The project Implementer will be responsible for managing household data collection and entry into the project database. Field personnel will receive training on how to properly deal with surveying techniques and reduce errors. The project database will record the start and end dates of each monitoring period and record the emission reductions attributable to each monitoring period.

Appropriate record keeping procedures will be implemented to ensure that each monitoring period data set can be transparently attributed to its corresponding project, preventing any occurrences of double counting. An internal review of the project database will be able to determine the status of project—the duration of previous monitoring periods, the households delivering monitoring data, and current verification activities.

Assessment for Leakage

According to methodology VMR0006, version 1.1, section 8.3; Leakage shall be considered as default 0.95 in accordance with Section 5.4 of AMS-IL.G. therefore default value of leakage as 0.95 will be applied.

Monitoring Reporting

The PP will assess all monitoring data and produce one monitoring report for the project for the VVB to verify corresponding to the preceding monitoring period the project. This report will present the data relating to the emission reductions generated by the project at the time of the monitoring period.

APPENDIX 1: LOCAL STAKEHOLDER CONSULTATION

1. Newspaper Notice posted on a national newspaper “Civitas” in Lao PDR.




Local Stakeholder Consultation Meeting

“Installation of High Efficiency Wood Burning Cookstoves in Lao PDR”

C-Quest Capital Stoves Asia Ltd is developing a Project Activity (PA) titled “Installation of High Efficiency Wood Burning Cookstoves in Lao PDR” and is seeking its registration with VERRA as Verified Carbon Standard Project. The PA aims to disseminate Improved Biomass Cookstoves in Lao PDR as a replacement of three stone fires or traditional pot supports stoves and thereby resulting in reduction of exposure to air pollution caused due to inefficient burning of biomass fuels and the depletion of forest cover caused by unsustainable harvesting of firewood. In turn this will lead to reduced global greenhouse gas emissions on account of less quantity of biomass burnt.

In this regard CQC extends an invitation to all stakeholders as well as interested parties to join us for a live **Webinar on Thursday, April 28, 2021, at 10:00 AM (Vientiane Local Time)** to discuss project design and its effect on sustainable development as well as to give your valuable feedback.

For registration for webinar please contact us on +91 9818031981 (WhatsApp)
Or write to us at: info@cquestcapital.com
Or visit : <https://cquestcapital.com/blogs/news-updates>

You may also send in your queries or comments at the above-mentioned address.

ກອງປະຊຸມປຶກສາຫາລືບັນດາຫຸ້ນສ່ວນໃນທ້ອງຖິ່ນ

“ການຕິດຕັ້ງເຕົາປະຍັດທີ່ໃຊ້ໄມ້ຟື້ນທີ່ມີປະສິດທິພາບສູງໃນ ສປປ ລາວ”

ບໍລິສັດ C-Quest Capital Stoves Asia Ltd. ກຳລັງພັດທະນາກິດຈະກຳໂຄງການ (PA) ພາຍໃຕ້ຫົວຂໍ້ “ການຕິດຕັ້ງເຕົາປະຍັດທີ່ໃຊ້ໄມ້ຟື້ນທີ່ມີປະສິດທິພາບສູງໃນ ສປປ ລາວ” ແລະ ກຳລັງດຳເນີນການສຳຄັນທາງການກັບ VERRA ເຊິ່ງເປັນໂຄງການສັບສອນມາດຕະຖານຄາບອນ. PA ມີເປົ້າໝາຍໃນການແຈກຢາຍເຕົາປະຍັດຊີວະມວນໃນ ສປປ ລາວ ເພື່ອຫຼຸດຜ່ອນເຕົາຟື້ນສາມລ້າງ ຫຼື ເຕົາໄຟແບບດັ້ງເດີມ ແລະ ຍັງສົ່ງຜົນສົນທິໃນການຫຼຸດຜ່ອນການສຳຜັດໄມ້ລະພິດທາງອາກາດ ທັນເບື້ອງມາຈາກການເຂົ້າໄຊເຊື້ອໄຟຊີວະມວນທີ່ໄດ້ປະສິດທິພາບ ແລະ ຊ່ວຍຊັບຊ້ອນການຫຼຸດລົງຂອງພື້ນທີ່ປົກຄຸມປ່າໄມ້ເບື້ອງຈາກການຊອກຫາໄມ້ຟື້ນທີ່ສາດຄວາມບັນຍັດ. ພ້ອມດຽວກັນນັ້ນກໍ່ຍັງເປັນການຫຼຸດຜ່ອນການປ່ອຍອາກາດອາຍພິດເຮືອນແກ້ວໃນທົ່ວໂລກ ເນື່ອງຈາກປະສິດທິພາບການເຂົ້າໄຊເຊື້ອໄຟຊີວະມວນມີໜ້ອຍລົງ.

ເພື່ອໃຫ້ບັນດາຜູ້ກ່ຽວຂ້ອງເຂົ້າຮ່ວມ CQC ສະຖານະການຊຸມຊົນບັນດາພາກສ່ວນທີ່ກ່ຽວຂ້ອງທຸກທ່ານ ຕະຫຼອດເຖິງບັນດາທ່ານທີ່ສົນໃຈເຂົ້າຮ່ວມ ງານສຳມະນາທາງໄກ ເຊິ່ງຈະຈັດຂຶ້ນໃນວັນພະຫັດ, ທີ 28 ເມສາ 2021, ເວລາ 10:00 ໂມງເຊົ້າ (ເວລາທ້ອງຖິ່ນນະຄອນຫຼວງວຽງຈັນ) ເພື່ອປຶກສາຫາລືກ່ຽວກັບການອອກແບບໂຄງການ ແລະ ຜົນກະທົບຕໍ່ການພັດທະນາແບບຍືນຍົງ ຕະຫຼອດເຖິງຂໍ້ສະເໜີແນະນຳມີຄວາມໝາຍຄວາມສຳຄັນຈາກທຸກທ່ານ.

ສຳລັບການສົງຄະນະບຸນເຂົ້າຮ່ວມງານສຳມະນາທາງໄກ ກະລຸນາຕິດຕໍ່ພວກເຮົາທີ່ +91 9818031981 (WhatsApp)
ຫຼື ສູນຈົດໝາຍທາງພວກເຮົາໄດ້ທີ່: info@cquestcapital.com
ຫຼື ເຂົ້າເບິ່ງຂໍ້ມູນເພີ່ມເຕີມທີ່ໄວ້ບໍ່ໄຊ: <https://cquestcapital.com/blogs/news-updates>

ທ່ານສາມາດສົງຄຳຖາມ ຫຼື ຄຳຄິດເຢັນຕາມທີ່ຢູ່ທີ່ໄດ້ລະບຸໄວ້ລຳດັບເທິງນີ້.

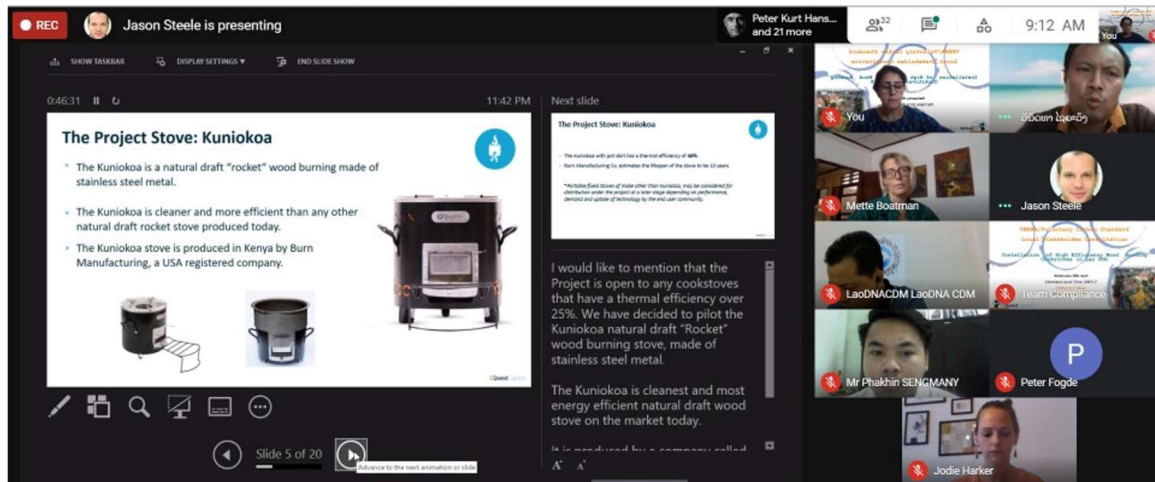
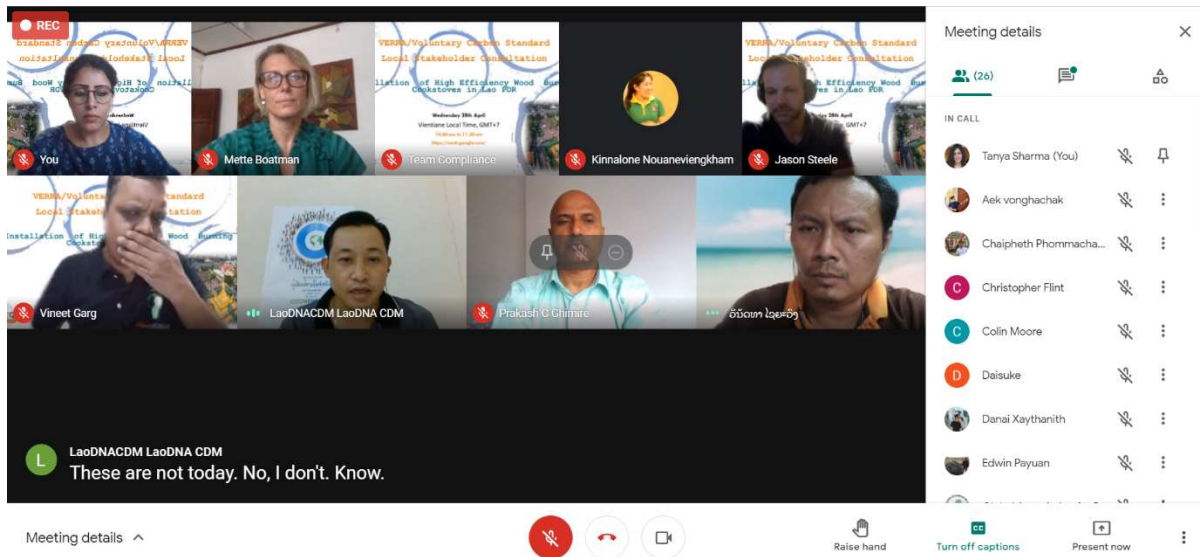
59

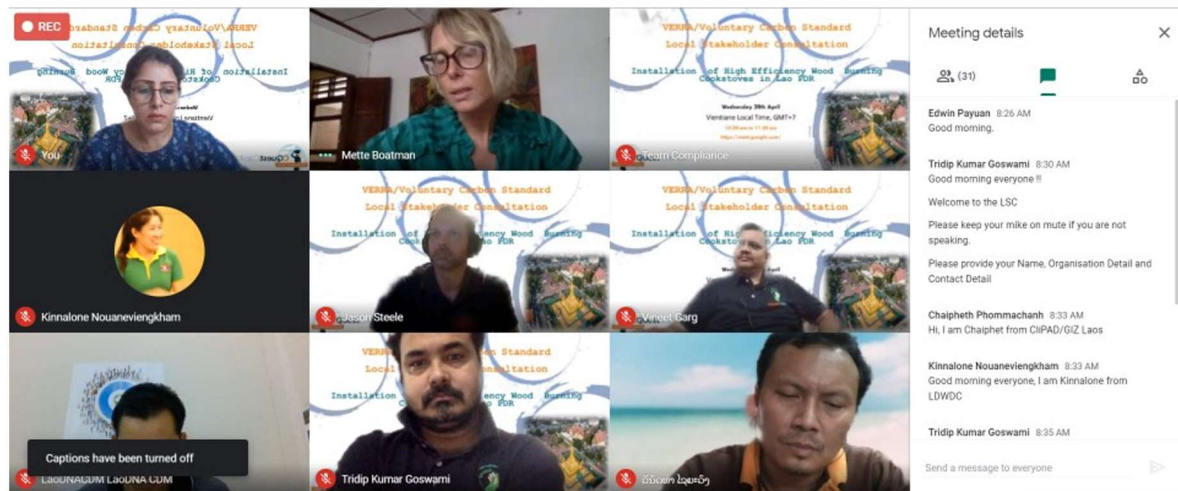
3.Stakeholders Meeting Question and Answers

Name	Stakeholder Comment	Response by CQC
Cristopher Flint	How will the new project decrease the pressure on forests and reduce levels of deforestation?	Our projects undertake monitoring on a sample basis across the population of the households to meet a certain statistical significance. It has been estimated that ~3 tons of CO2 saved/stove/year which is replicated over the entire population to estimate the impacts of the project activity.
Nelson Gapare	What are the major risk factors that have been identified during the implementation and assessment of the project?	User response is one of the greatest risks associated with project implementation as it determines the acceptability and longevity of project implementation. It is important to understand the factors that influence the choices of community and end users to continue to use the improved cookstoves and avoid the discontinuation of its usage. For the same consistent effort is made in terms of providing these cookstoves at low costs and undertaking regular monitoring to ensure its efficient working and continuous adoption among the communities.
Nelson Gapare	Does the project propagates the production of cookstoves in Laos and how does it fit in building capacity of the local cookstove value chain in Laos?	The project proponents have long been trying to find the best technology that suits best to the local context which has led us to importing these stoves into Laos in our current project. We also tried to install locally produced stoves in one of our earlier projects, but it did not last for more than a year. The key to the adoption and continuation of these projects lies in the quality of the stove, which is somehow compromised in the local made products.
: Global Association for People and the Environment	Are you working with ARMI- Association for rural mobilization and improvement which is working in the field of rural development in Laos. Also, what are your plans of working with other NGOs and partnering with them or organizations in the upcoming projects?	We are aware about ARMI and the work done by the organization in the past with us in projects, but this organization has not worked on firewood stoves in rural areas.
Chaipheth Phommachanh	He provided an information about their project called Climate	We are very open to any kind of association.

	Protection to avoid deforestation at GIZ, Laos. They are supporting the Laos government, especially Ministry of Agriculture and Forestry and Ministry of Natural Resources and Environment in implementing the emission reduction program and supporting the local administration or villagers to manage the forests. They are also trying to support their livelihood and suggested that the initiative of CQC supports their main objective, and we are ready to support the project and initiatives at Laos though not financially but technically	
Kinnalone	Ms. Kinnalone is from the Laos Disabled Women's Development Centre which is working on empowering and helping women through trainings, employment and capacity building. The organization has been receiving support in different dimensions from other organizations since a long time and the center is willing to collaborate with international organizations and other development partners in Laos as the mandate of the Centre is to support the women member living in the rural areas and to build the leadership skills and promote the gender equality among not only the women but also the male counterparts.	

4.Stakeholder Consultation Webinar Photos





Meeting details

(31)

Edwin Payuan 8:26 AM
Good morning.

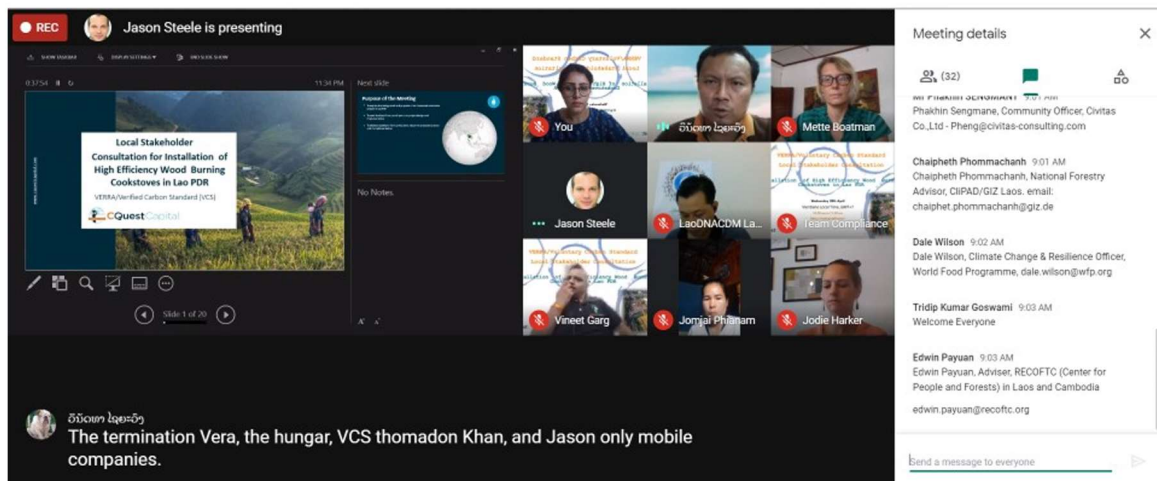
Tridip Kumar Goswami 8:30 AM
Good morning everyone !!
Welcome to the LSC
Please keep your mike on mute if you are not speaking
Please provide your Name, Organisation Detail and Contact Detail

Chaiapheth Phommachanh 8:33 AM
Hi, I am Chaiaphet from CIIPAD/GIZ Laos

Kinnalane Nouaneviengkham 8:33 AM
Good morning everyone, I am Kinnalane from LDWDC

Tridip Kumar Goswami 8:35 AM

Send a message to everyone



Meeting details

(32)

Phakhin Sengmane, Community Officer, Civitas Co.,Ltd - Pheng@civitas-consulting.com

Chaiapheth Phommachanh 9:01 AM
Chaiapheth Phommachanh, National Forestry Advisor, CIIPAD/GIZ Laos. email: chaiaphet.phommachanh@giz.de

Dale Wilson 9:02 AM
Dale Wilson, Climate Change & Resilience Officer, World Food Programme, dale.wilson@wfp.org

Tridip Kumar Goswami 9:03 AM
Welcome Everyone

Edwin Payuan 9:03 AM
Edwin Payuan, Adviser, RECOFTC (Center for People and Forests) in Laos and Cambodia
edwin.payuan@recoftc.org

Send a message to everyone

Local Stakeholder Consultation for Installation of High Efficiency Wood Burning Cookstoves in Lao PDR
VERA/Voluntary Carbon Standard (VCS)
CCQuest Capital

Slide 1 of 20

Agenda of the Meeting

No Notes

Termination Vera, the hungar, VCS thomadon Khan, and Jason only mobile companies.

5. List of people who attended the Stakeholder Meeting

NAME	Email
Akhilesh Joshi	compliance@cqcbrighsparks.in
Saminder Bedi	sami*****@***.com
Tanya Sharma	tany*****@***.com
Jason Steele	jaso*****@***.com
Tridip Kumar Goswami	trid*****@***.com
Mette Boatman	mett*****@***.com
Vineet Garg	vkg.*****@***.com
Pooja Verma	p.a.*****@***.com
Phonesavanh Chindamany	phon*****@***.com
Christopher Flint	chri*****@***.com
Peter Fogde	pete*@***.com
Nelson Gapare	gapa***@***.com
Aek Garnett	aekg*****@***.com
Prakash C Ghimire	prak*****@***.com
Peter Kurt Hansen	pkh*****@***.com
Jodie Harker	hark*****@***.com
Mr. Syamphone Sengchandala	laoc*****@***.com
Colin Moore	coli*****@***.com
Kinnalone Nouaneviengkham	kinn*****@***.com
Edwin Payuan	edwi*****@***.org
Global Association for People and the Environment	gape*****@***.com
Jomjai Phianam	jomj**@***.com
Chaipheth Phommachanh	pmcc*****@***.com
POPZ Phongthanou	phon*****@***.com
Tahirih Tamangen	tahi*****@***.org
Aek Vonghachak	aekr*****@***.com
Dale Wilson	smud*****@***.com
Danai Xaythanith	dxay*****@***.com
Vinattha Xayavong	vina*****@***.com
Daisuke	-
Mr Phakhin SENGMAHY	-
Phitthiphong	-
Mrs. Aruni Phommakon	-

6. Sample Evaluation Form

The sample evaluation form was presented to the Stakeholders in their local language.

ແບບຟອມການປະເມີນ (ຂໍ້ສະເໜີແນະ)

1. ຊື່*	
2. ຕຳແໜ່ງ	
3. ຊື່ອົງການຈັດຕັ້ງ	
4. Email*	
5. ທ່ານປະທັບໃຈຫຍັງແດ່ກ່ຽວກັບການດຳເນີນກິດຈະກຳໂຄງການ (PA)?	
ຈະຊ່ວຍໃຫ້ສະມາຊິກໃນຄົວເຮືອນສຳຜັດກັບຄວາມພິດຈາກການປຸງແຕ່ງອາຫານໜ້ອຍລົງ	☐
ຈະຊົມໃຊ້ໄມ້ຟືນໜ້ອຍລົງ	☐
ຈະຊ່ວຍປະຢັດເວລາໃນການປຸງແຕ່ງອາຫານ ແລະ ການເກັບຟືນ	☐
ບໍ່ມັກຫຍັງເລີຍກ່ຽວກັບໂຄງການ	☐
ອື່ນໆ (ກະລຸນາລະບຸ) -	
6. ທ່ານຄິດວ່າໂຄງການນີ້ຈະເປັນປະໂຫຍດສຳລັບຜູ້ຊົມໃຊ້ ຫຼື ບໍ່?	
ແມ່ນ	☐
ບໍ່	☐
ອື່ນໆ (ກະລຸນາລະບຸ)	
7. ທ່ານຄິດວ່າໂຄງການນີ້ຈະເປັນປະໂຫຍດຕໍ່ສິ່ງແວດລ້ອມ ຫຼື ບໍ່?	
ແມ່ນ	☐
ບໍ່	☐
ອື່ນໆ (ກະລຸນາລະບຸ)	
8. ທ່ານຄິດວ່າການນຳໃຊ້ເຕົາຂອງໂຄງການຈະສົ່ງຜົນດີຕໍ່ສຸຂະພາບຂອງແມ່ຍິງ ແລະ ເດັກນ້ອຍ ຫຼື ບໍ່?	
ແມ່ນ	☐
ບໍ່	☐
ອື່ນໆ (ກະລຸນາລະບຸ)	
9. ທ່ານຕ້ອງການມີສ່ວນຮ່ວມກັບໂຄງການນີ້ ຫຼື ບໍ່ ຫາກມີໂອກາດ?	
ແມ່ນ	☐
ບໍ່	☐
ອື່ນໆ (ກະລຸນາລະບຸ)	
10. ທ່ານມີຂໍ້ສະເໜີແນະກ່ຽວກັບການອອກແບບເຕົາປຸງແຕ່ງອາຫານ/ໂຄງການ ຫຼື ບໍ່?	

11. ທ່ານຮູ້ກ່ຽວກັບໂຄງການອື່ນໆທີ່ກຳລັງດຳເນີນການຢູ່ ແລະ/ ຫຼື ຕາມແຜນ (ກົດຈະກຳຕ່າງໆຂອງໂຄງການໃດໜຶ່ງ/ກົດຈະກຳໂຄງການ) ພາຍໃຕ້ກົນໄກ GHG ໃນເຂດທ້ອງຖິ່ນຂອງທ່ານ ຫຼື ບໍ່? ຖ້າຮູ້, ທ່ານສາມາດໃຫ້ລາຍລະອຽດໄດ້ ຫຼື ບໍ່?
12. ທ່ານຮັບຮູ້ກ່ຽວກັບຜົນກະທົບທາງລົບຈາກໂຄງການ ຫຼື ບໍ່? ຖ້າຮູ້, ທ່ານສາມາດໃຫ້ລາຍລະອຽດໄດ້ ຫຼື ບໍ່?
13. ຂໍສະເໜີແນະນຳອື່ນໆ

APPENDIX 2: BASELINE FUEL CONSUMPTION

For assessment of the baseline consumption the approach is as followed:

As per applied methodology VMR0006 Sec 8.4 (pg10) the value of baseline fuel consumption (B_{old}) can be sourced from historical data or baseline surveys. Alternatively, a default value of 0.5t/capita/year may be used.

PP chosen the historic data option for the estimation of baseline fuelwood consumption which is based on the historical data from available literature ³⁴. The value is calculated for comparison for comparison of baseline and project case calculation.

PP would monitor the $B_{y=1,new,il,j,survey}$ in the first year of monitoring period inline with the applied methodology VMR0006 ver 1.1 as described in sec 5.3 of the VCS PD monitoring plan. The ex-post calculation of ERs will be done based on survey results of grouped project conducted as detailed in section 4.4 & 5.3 of the VCS PD.

Calculation of baseline fuelwood consumption

Table 10: Estimation of baseline fuelwood consumption

S.No	Parameter	Unit	Value	Source
A	Fuel wood consumption by Household (Yr. 2022)	m ³	3,560,000	UN Statistics Division data ³⁵
B	Density of wood	tons/ m ³	0.725	FAO [2]
C	Fuel wood consumption by households (Yr2022)	tons	2,581,000	Calculated (A*B)
D	Population of Lao PDR (Yr. 2022)	#	7,529,475	World Bank (Population data) ³⁶
E	Average household size	#	5.3	Lao PDR Population & Housing census ³⁷
F	Total number of Households in Lao PDR (Yr2022)	#	1,420,656	Calculated (D/E)
G	Share of households using wood for cooking	%	69.00%	Energy statistic of Lao PDR for yr. 2021, table 7.5 (pg. 112) ³⁸
H	Households using fuel wood for cooking	#	980,252	Calculated (F*G)
I	Baseline fuelwood consumption per household	tons /HH/yr.	2.633	Calculated (C/H)
J	Per capita fuel wood consumption	tons /capita/yr.	0.50	Calculated (I/E)

³⁴ The calculations sheet provided to VVB as provided in the Appendix 2 of the VCS PD

³⁵ https://data.un.org/Data.aspx?d=EDATA&f=cmID%3aFW%3btrID%3a1231#f_1

³⁶ <https://data.worldbank.org/indicator/SP.POP.TOTL?locations=LA>

³⁷ [Lao-Population-census-2015.pdf](#)

³⁸ [Energy statistic of Lao PDR \(2021\)](#)

Hence the Annual quantity of woody biomass used by improved cook stoves in tonnes (1.1769 ton/yr³⁹ as considered for ER estimates in this VCS PD⁴⁰), will not exceed the baseline fuelwood consumption (2.633 ton/yr as estimated in the Table 10 above in this VCS PD)

As more than one project stove may be required to achieve service levels equal to baseline stove. For such cases, the displaced biomass shall be apportioned between the project stoves while calculating baseline fuel consumption and following the item 8.4 of the applied methodology VMR0006 V 1.1 the apportioned fuel consumption arrives as follows:

Table 11: Apportioning of the baseline fuel consumption between the project stoves⁴¹

S.No	Parameter	Unit	Value	Source
1	Annual quantity of woody biomass that would have been used per person in the household in the absence of the project activity ($B_{old,p}$)	Tons/capita/Yr	0.50	Baseline fuelwood consumption assessment vide table 10 above
2	Average number of persons per HH (N_p HH)	Number	5.3	Baseline fuelwood consumption assessment table 10
3	Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices ($B_{old,HH}$)	Tons/HH/yr.	2.633	Calculated (S no 1* S.no 2)
4	No.of project devices per house hold ($N_{d,HH}$)	number	2	Project devices per HH (as per Sec 1.1 of the PDD)
5	Baseline fuel consumption apportioned per device ($B_{old,i,j}$)	Tons/device/yr	1.316	Calculated (S.no 3/S.no 4)

As per the apportioned fuelwood consumption estimated baseline fuelwood consumption vide table 10 & table 11 between the project stoves shows that. $B_{y=1,new,l,j,survey}$ will not exceed the baseline consumption⁴².

³⁹ Sum of the Fuel consumption considered for ex-ante estimation of both stoves ie 1.769t/yr = 1.0841 t/yr(SSM-32X)+0.6852t/yr(Naga stove) as each household will received up to 2 ICS.

⁴⁰ PP would monitor the $B_{y=1,new,l,j,survey}$ in the first year of monitoring period inline with the applied methodology VMR0006 ver 1.1 as described under sec 5.3 of this VCSPD

⁴¹ The equations used for calculating biomass consumed in absence of project activity in case more than one project stove is used in household as per applied methodology VMR0006 V1.1 (pg12)

⁴² Baseline fuel consumption apportioned per device is 1.316 tonnes/device /yr and the project devices (1.0841 t/yr(SSM-32X)+0.6852t/yr(Naga stove) is less than the per device fuel consumption